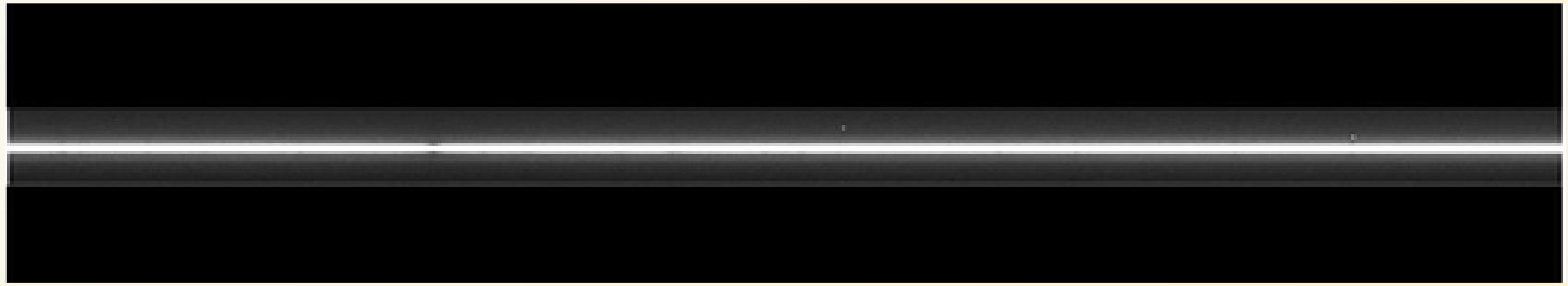
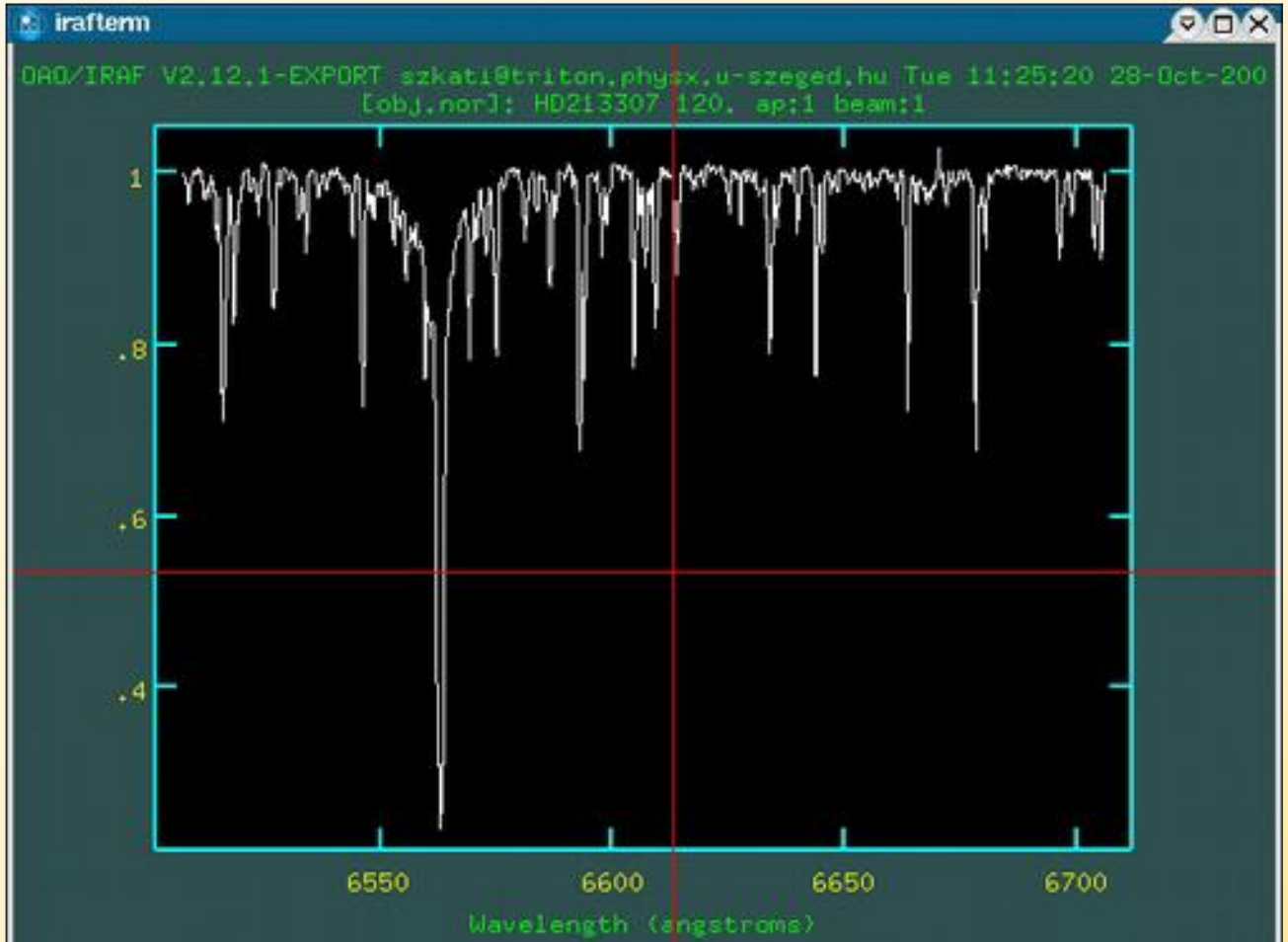
Reducing Cassegrain spectra

- Raw Spectrum.



- Continuum normalized and wavelength calibrated spectrum.



Reducing Cassegrain spectra

- Bias correction: to remove the readout noise.
- Cosmic ray subtraction (optional).
- Dark correction (optional): to remove noise due to the temperature of the CCD.
- Flat correction: to correct for the pixel-to-pixel variations.
- Extracting the aperture from the image.
- Sky subtraction (optional): to remove the sky emission lines.
- Wavelength calibration: determine the position CCD vs. wavelength correlation.
- Telluric correction (optional): to remove the sky absorption lines.
- Flux calibration (optional): to calculate accurate fluxes of the spectrum.
- Continuum normalization: to remove the continuum of the spectrum.
- Do science: cross-correlation, abundance analysis etc.

Listing the header info of fits files

- ls *fits >all.list
- images => util => epar imhead



=> bemenő file
=> kép kiterjesztés

Run with:

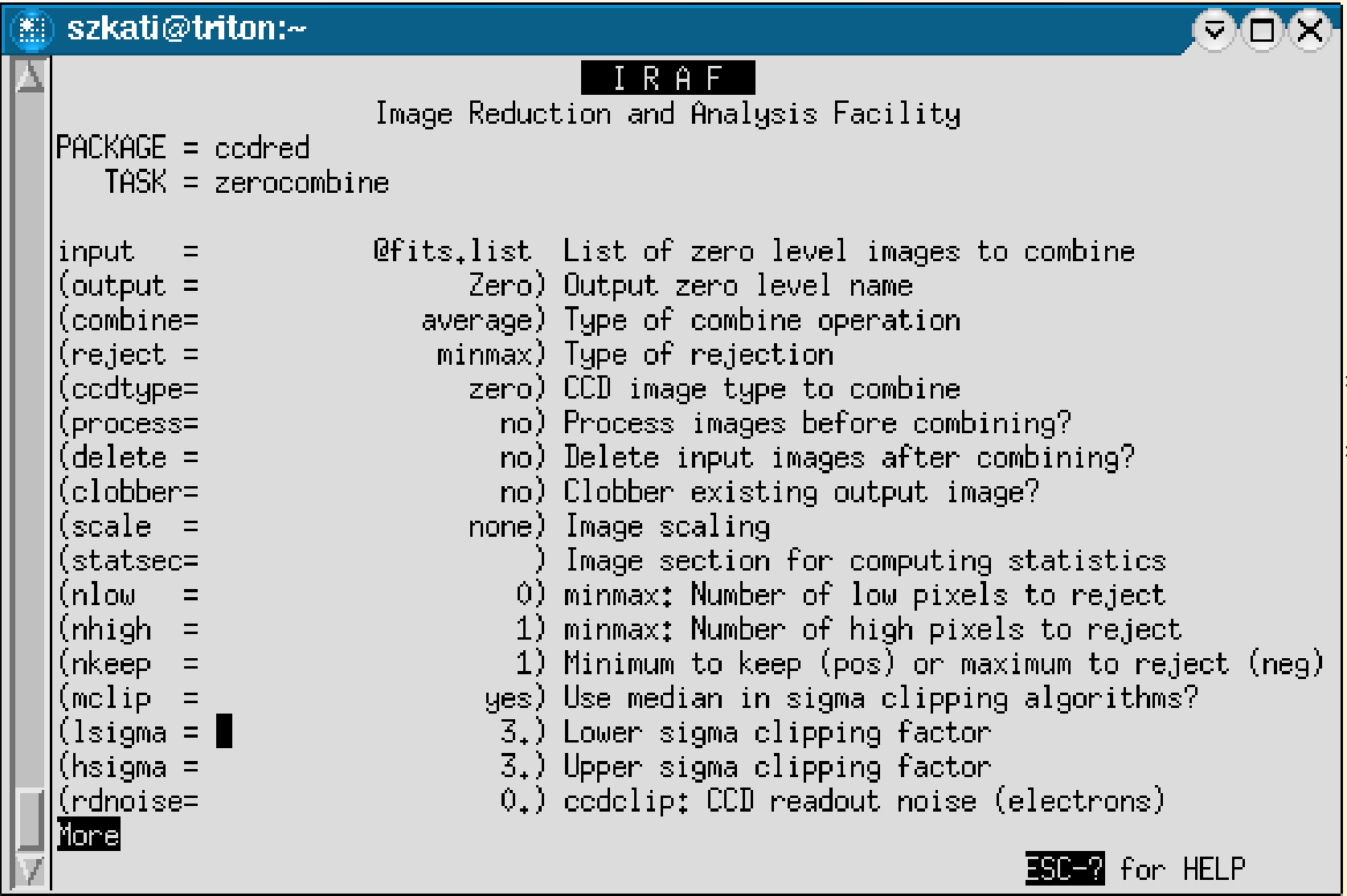
:g

Exit with:

:q

Bias correction: zerocombine

- noao => imred => ccdred => epar zerocombine



=> bemenő file, bias képek listája
=> kimenő file

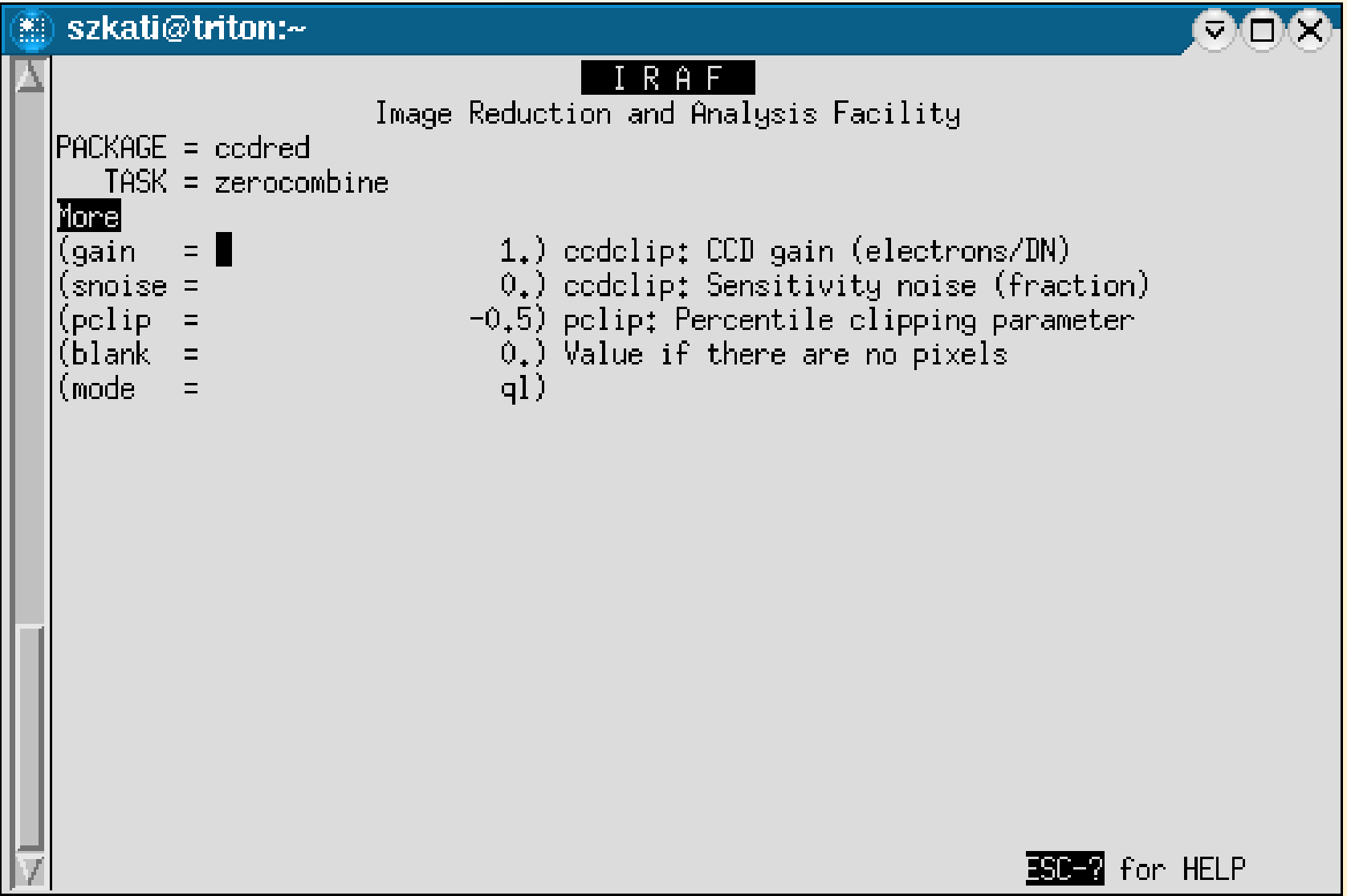
=> kép típusa

=> kombinált képek törlése

=> a CCD kiolvasási zaja

Bias correction: zerocombine

- noao => imred => ccdred => epar zerocombine



Run with:

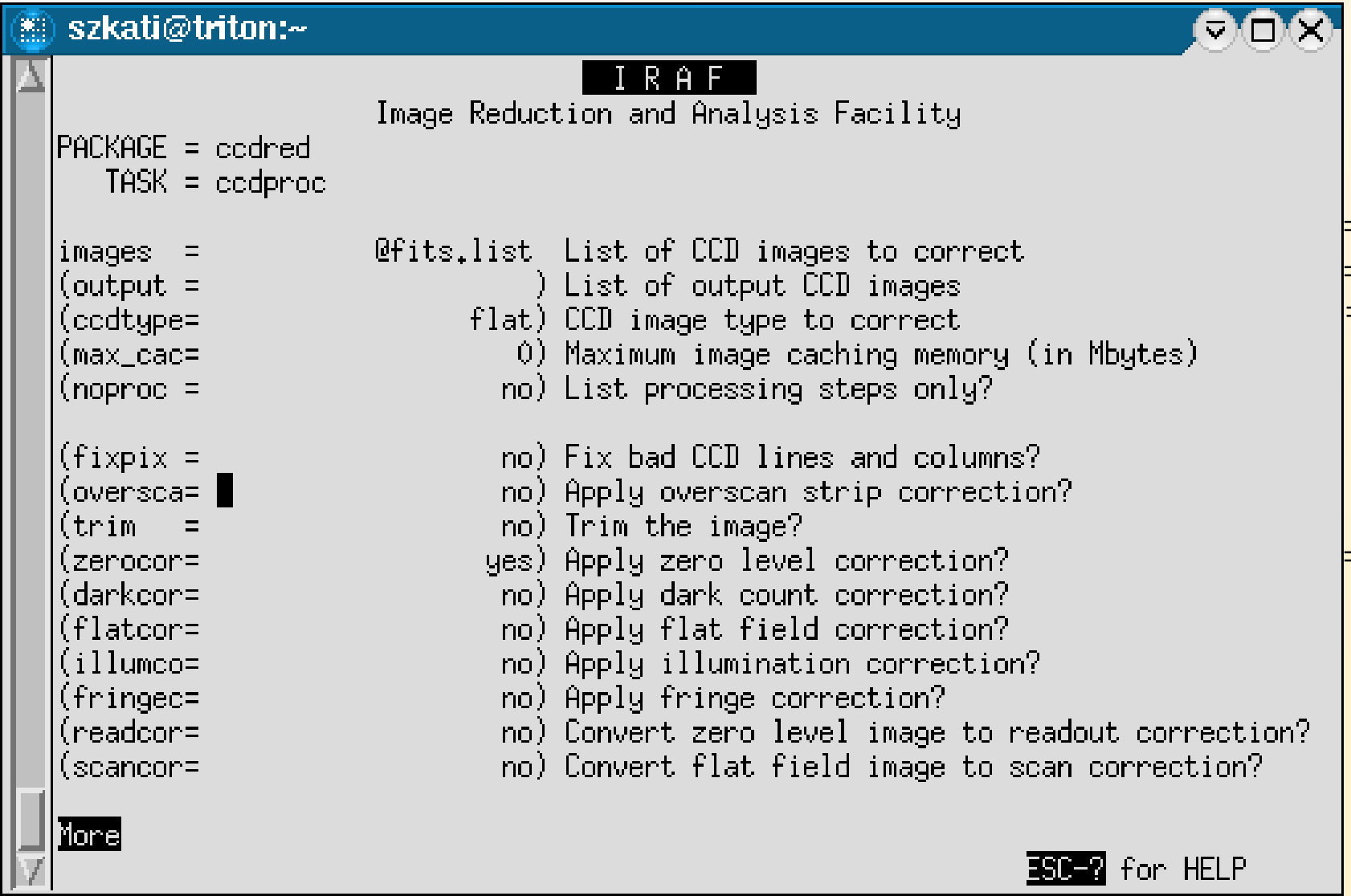
:g

Exit with:

:q

Bias correction: ccdproc

- noao => imred => ccdred => epar ccdproc

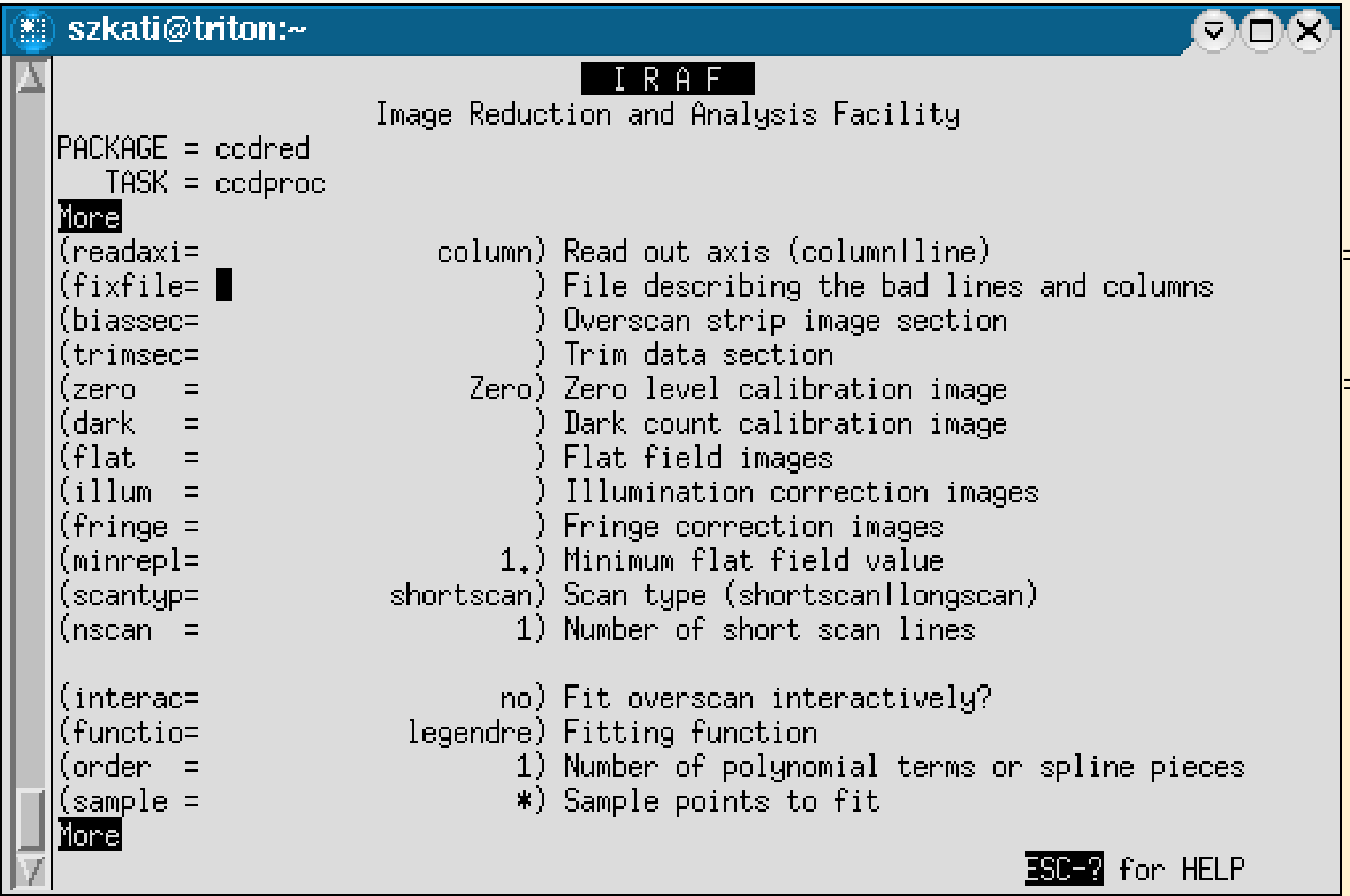


=> bemenő file
=> kimenő file
=> kép típusa (objects or flat)

=> bias korrekció

Bias correction: ccdproc

- noao => imred => ccdred => epar ccdproc

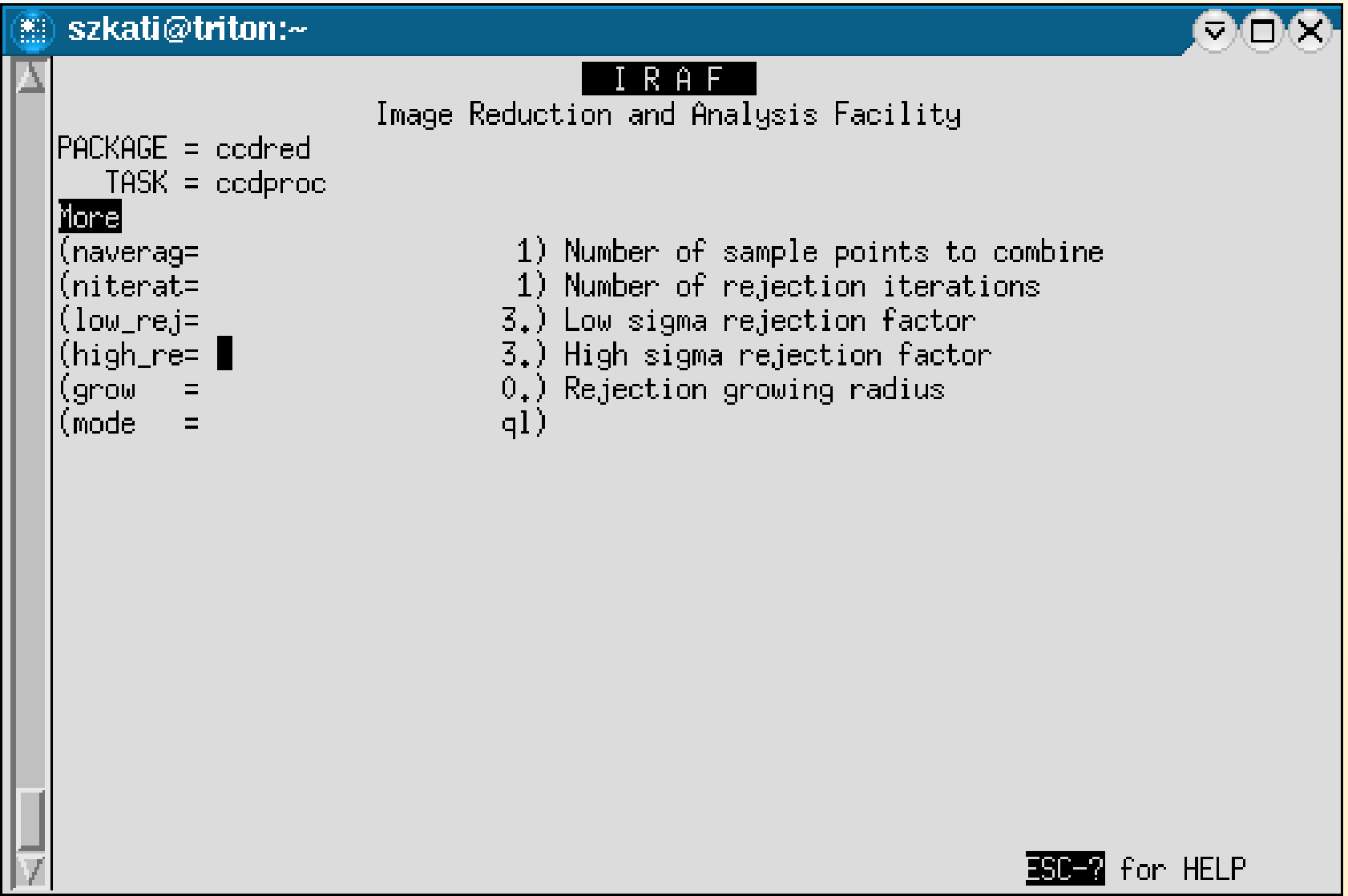


=> kiolvasási irány

=> biast korrigáló kép

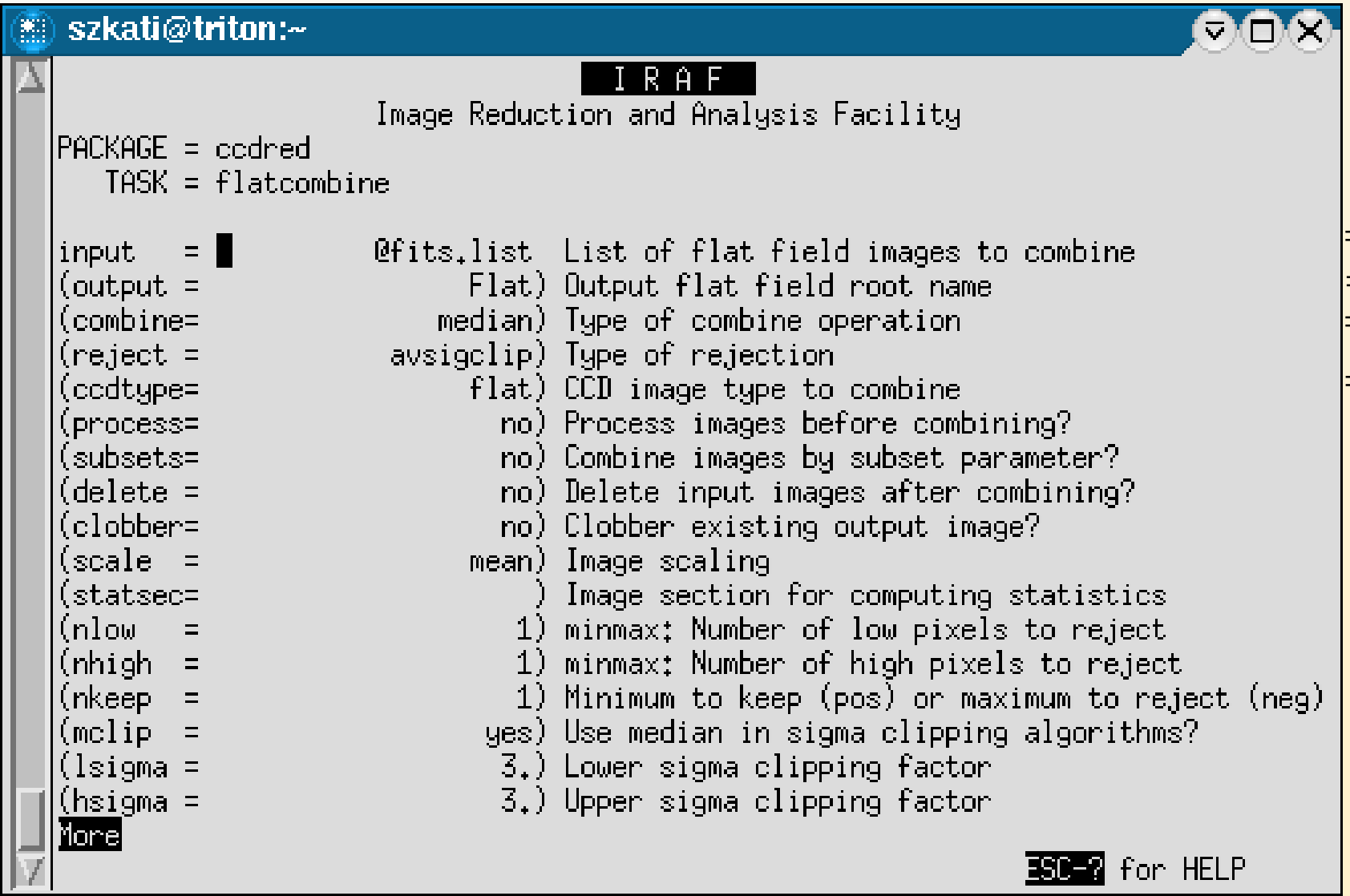
Bias correction: ccdproc

- noao => imred => ccdred => epar ccdproc



Flat correction: flatcombine

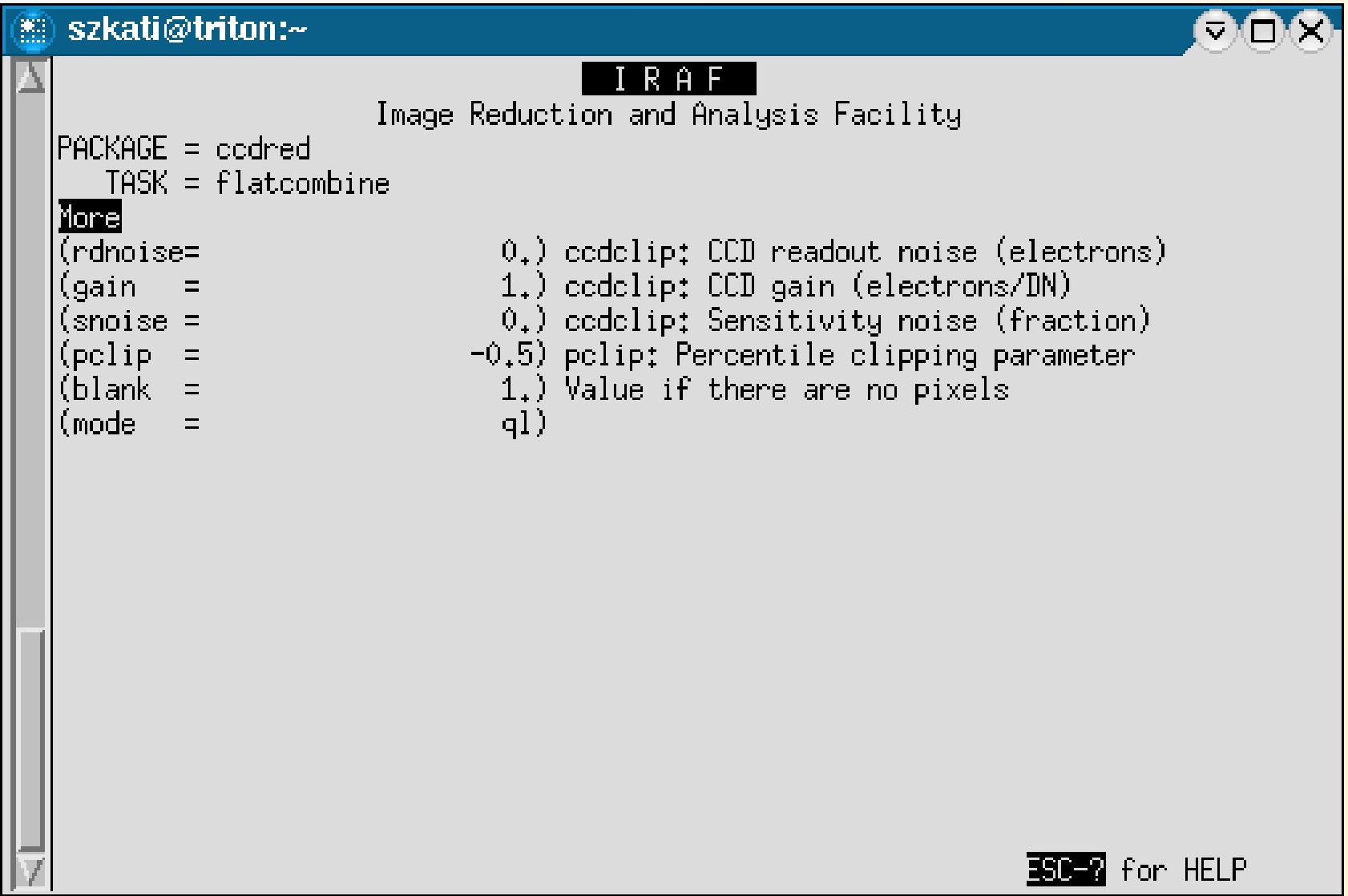
- noao => imred => ccdred => epar flatcombine



=> bemenő file (flat.list)
=> kimenő file
=> átlagolás típusa
=> kép típusa

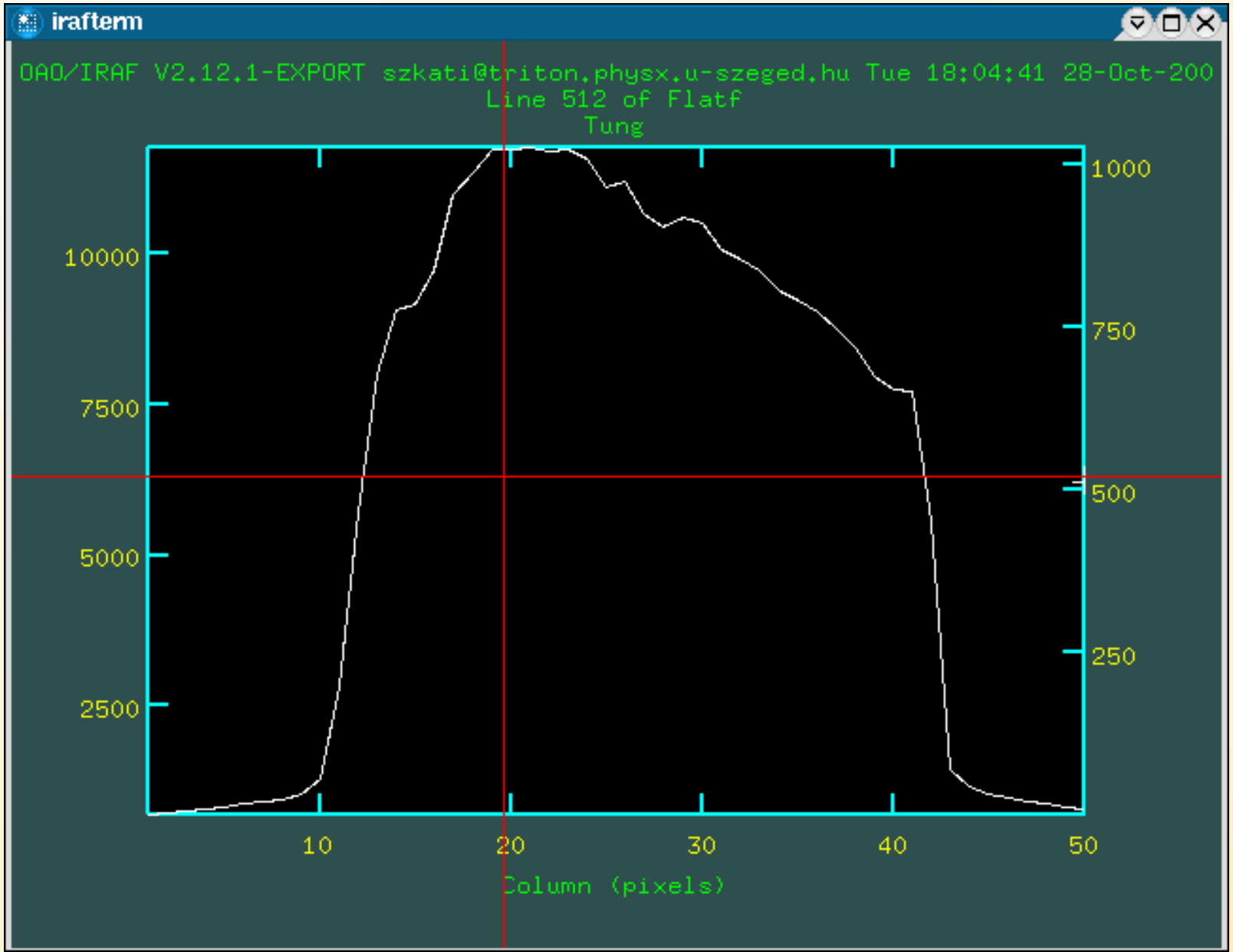
Flat correction: flatcombine

- noao => imred => ccdred => epar flatcombine



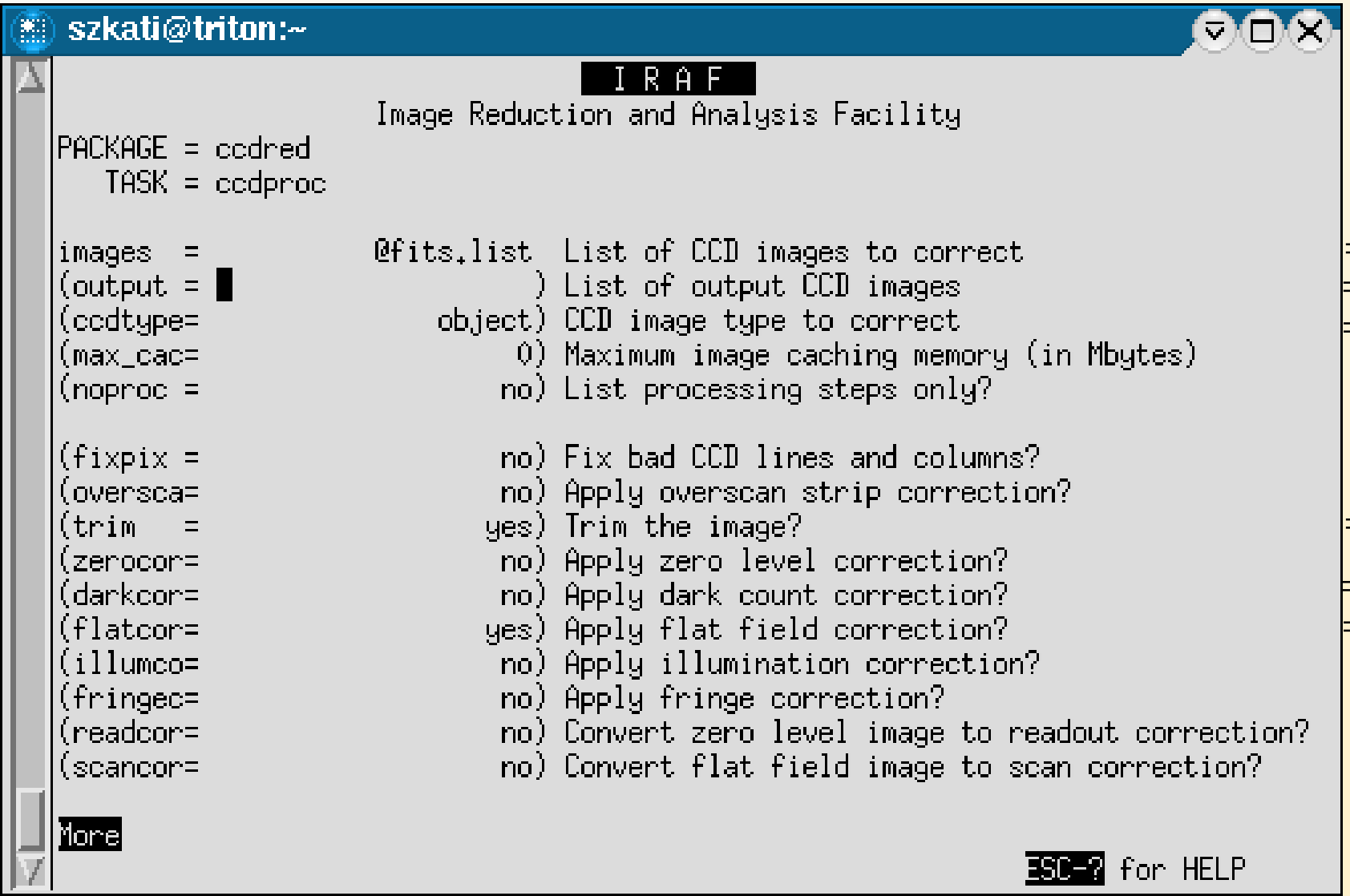
Flat correction: identify the usable region

- implot Flat.fits



Flat correction: ccdproc

- noao => imred => ccdred => epar ccdproc



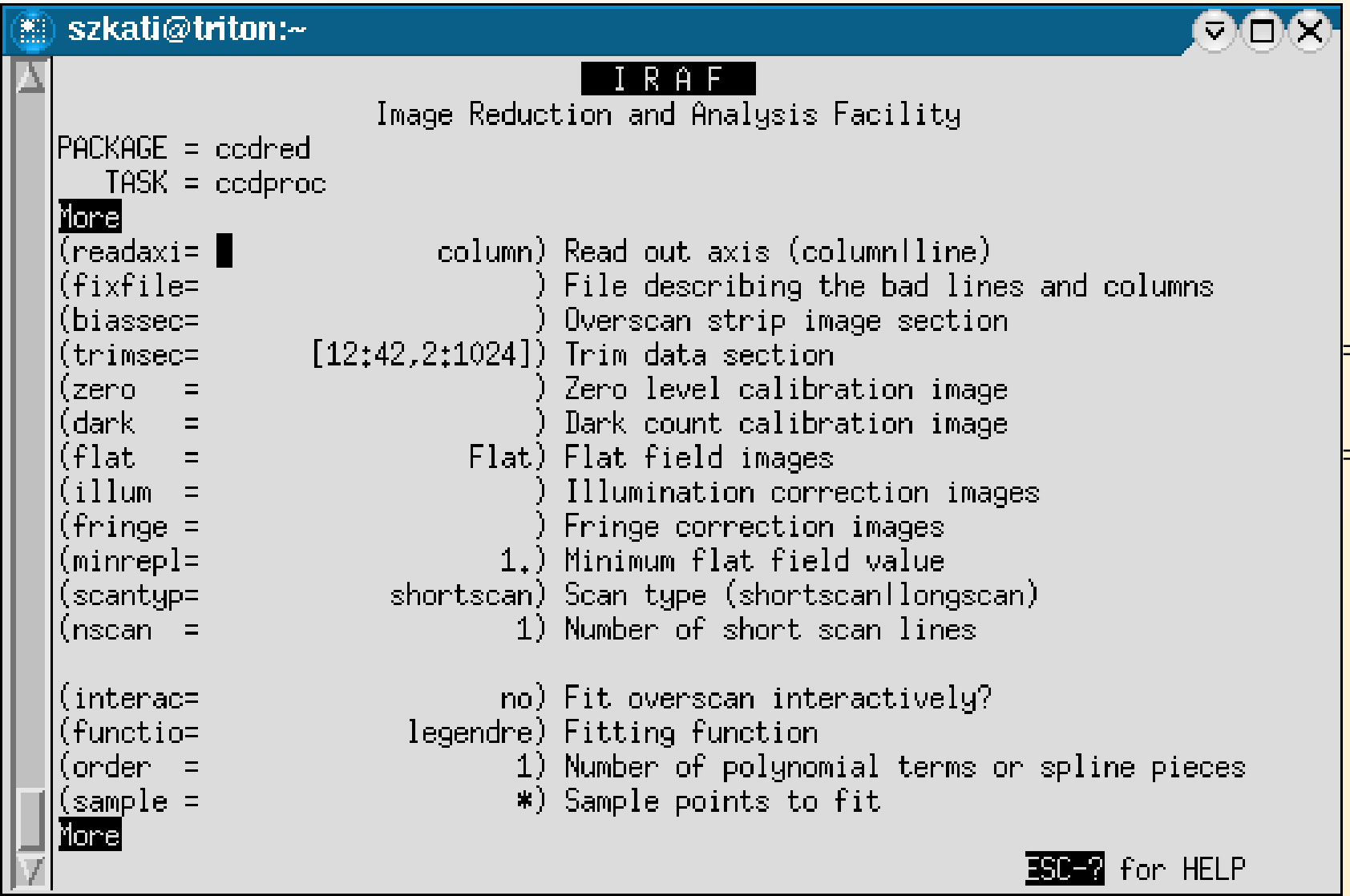
=> kép neve
=> kimenet kép neve
=> kép típusa

=> vágás

=> darkkorrekció
=> flatkorrekció

Flat correction: ccdproc

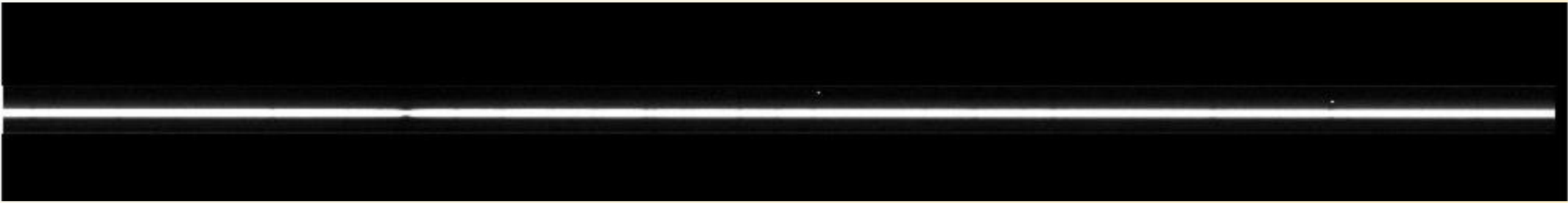
- noao => imred => ccdred => epar ccdproc



=> kivágandó régió

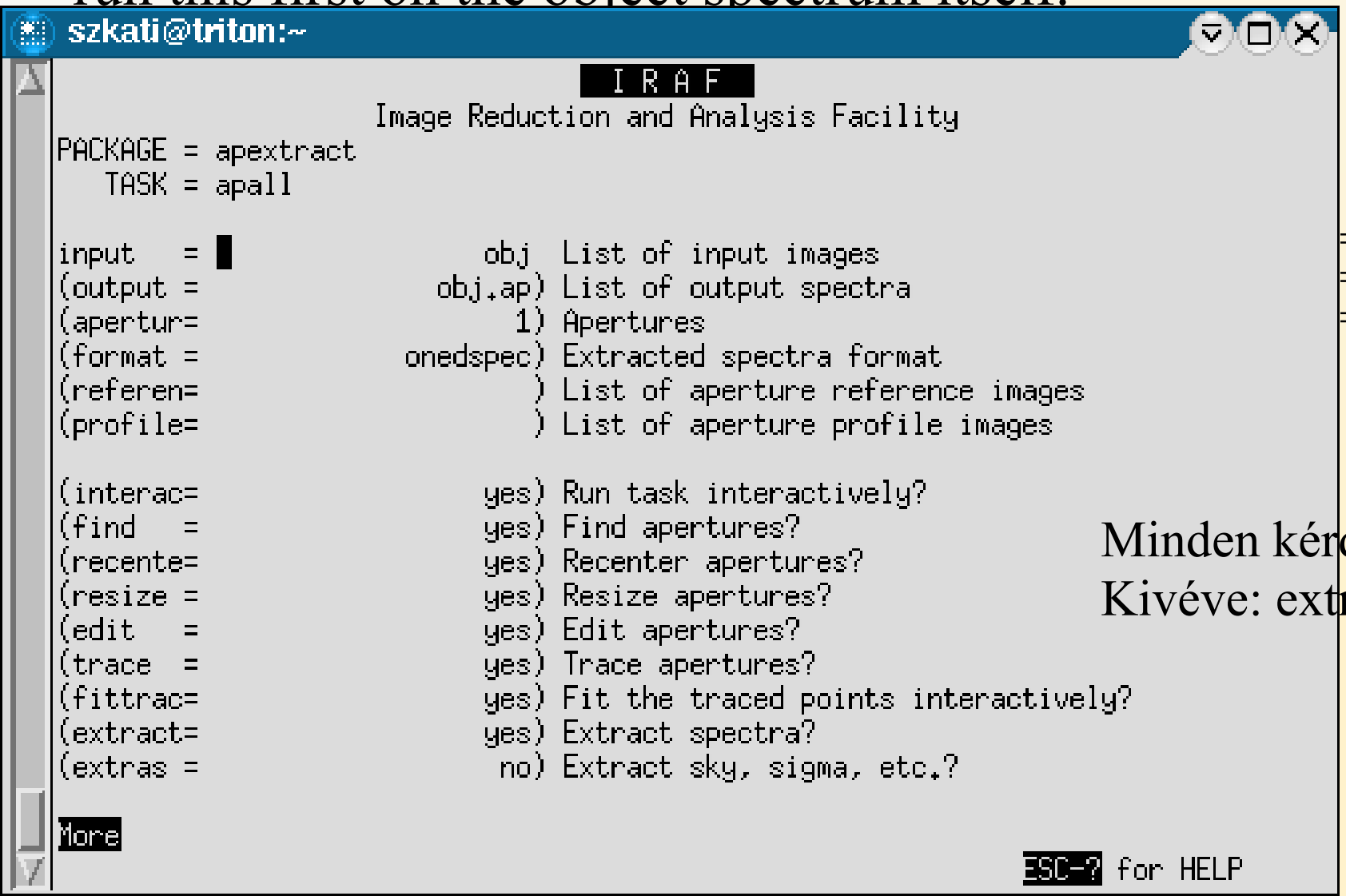
=> flat korrigáló kép

Flat correction: corrected image



Extracting the aperture: apall

- noao => twodspec => apextract => epar apall . It is important to run this first on the object spectrum itself.

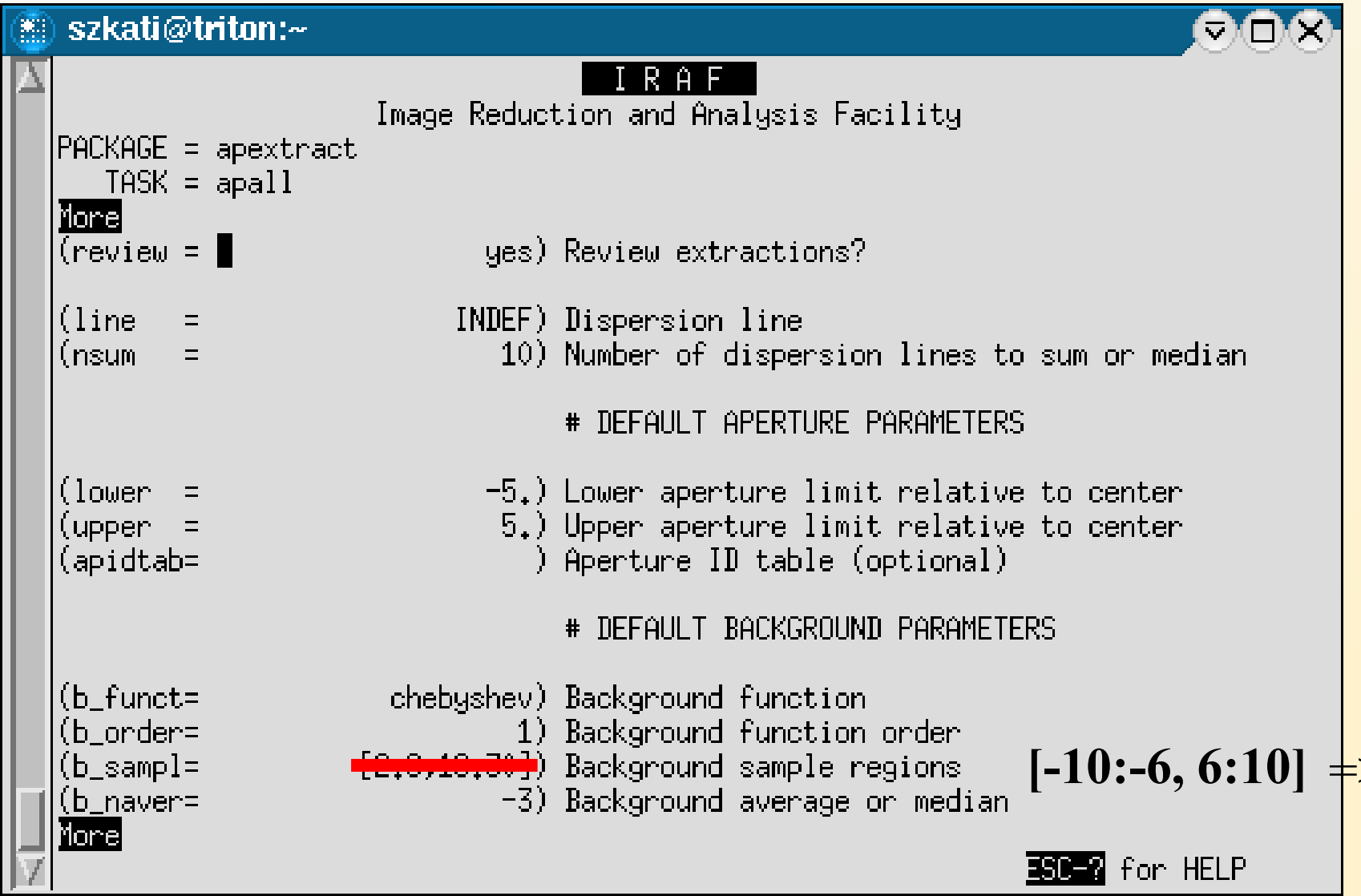


=> bemenő kép
=> kimenő kép
=> apertúrák száma

Minden kérdésre a válasz: yes
Kivéve: extras

Extracting the aperture: apall

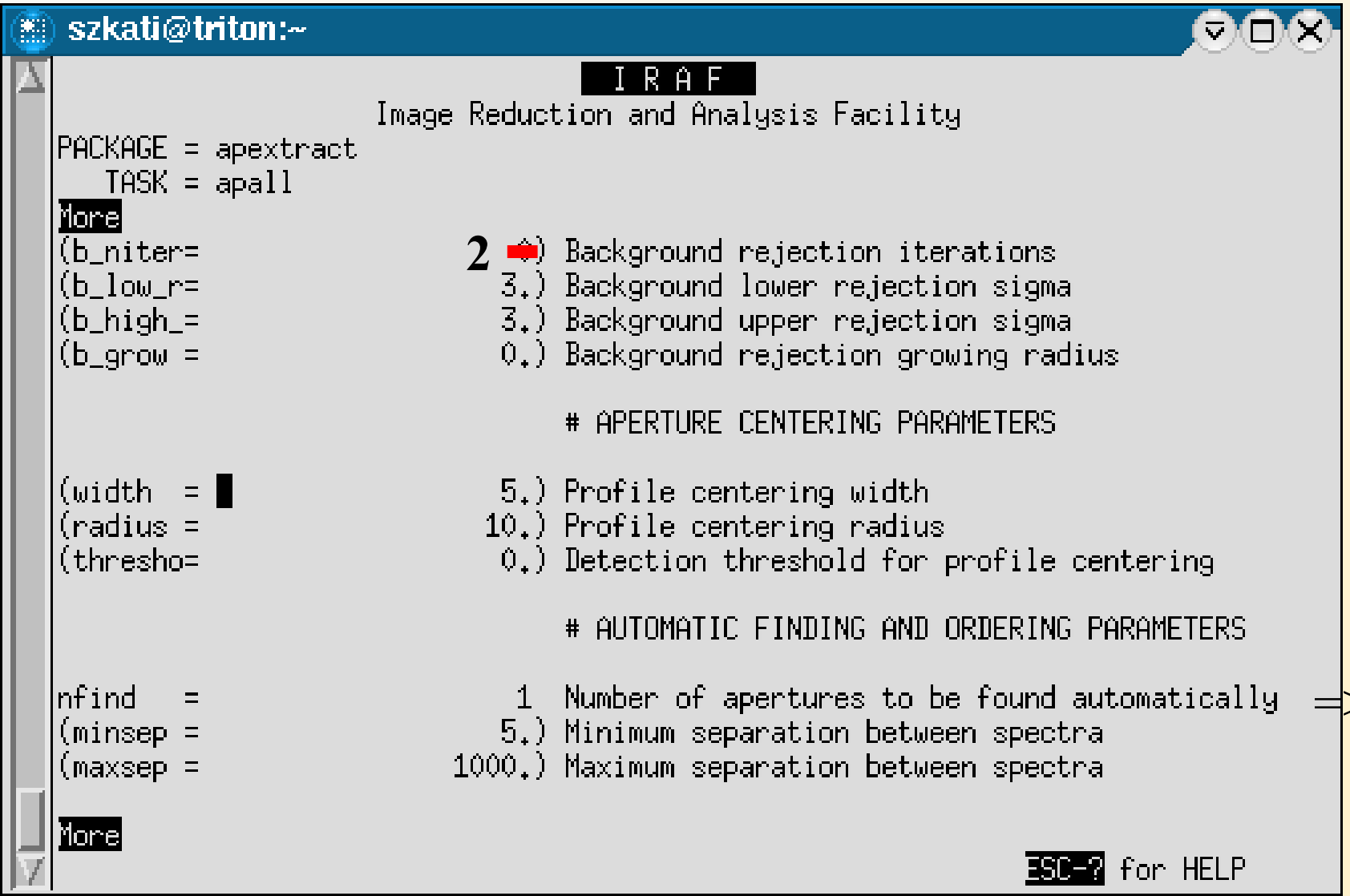
- noao => twodspec => apextract => epar apall



[-10:-6, 6:10] => háttér régió

Extracting the aperture: apall

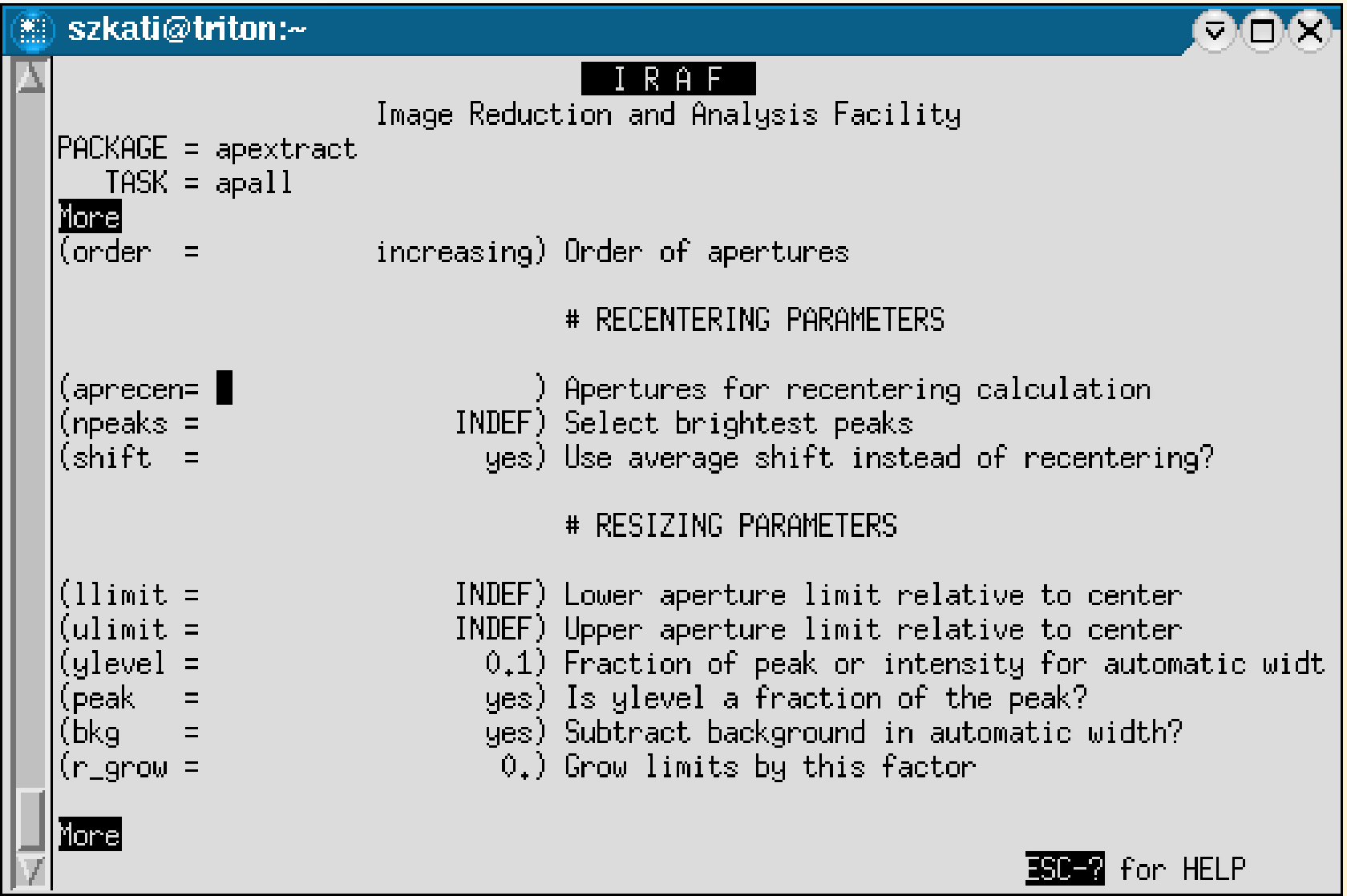
- noao => twodspec => apextract => epar apall



=> Aperturák száma

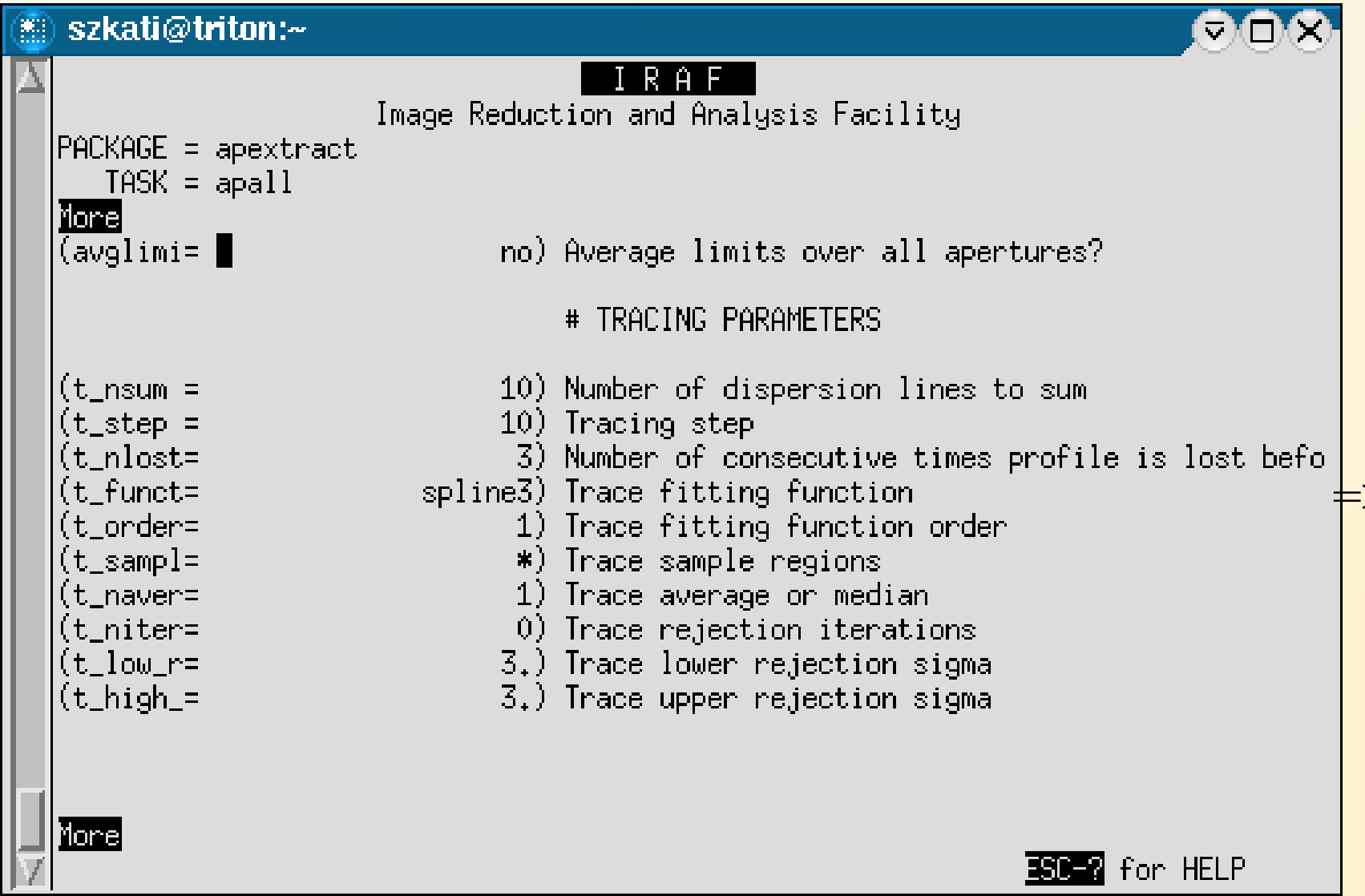
Extracting the aperture: apall

- noao => twodspec => apextract => epar apall



Extracting the aperture: apall

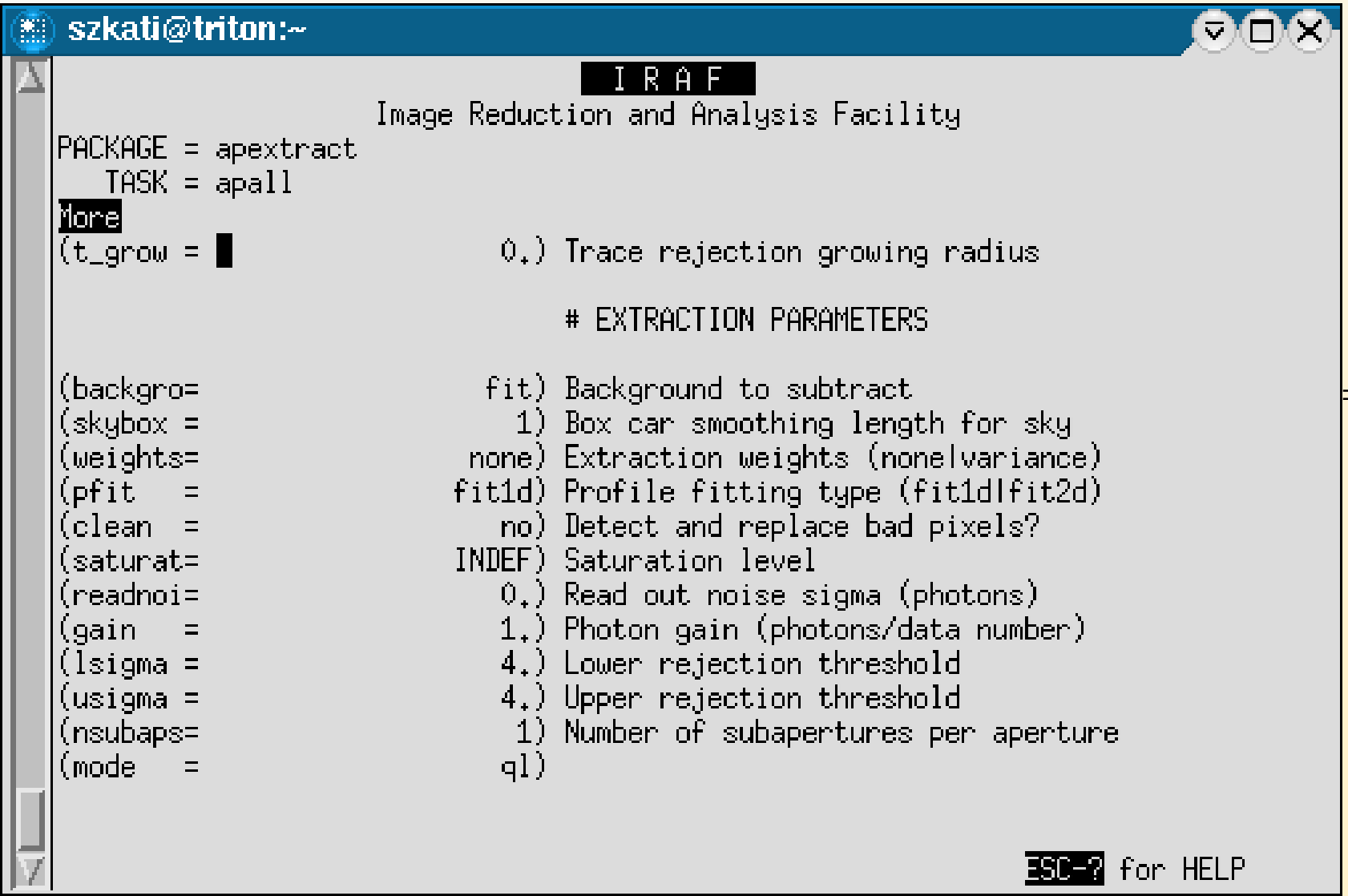
- noao => twodspec => apextract => epar apall



=> spline3

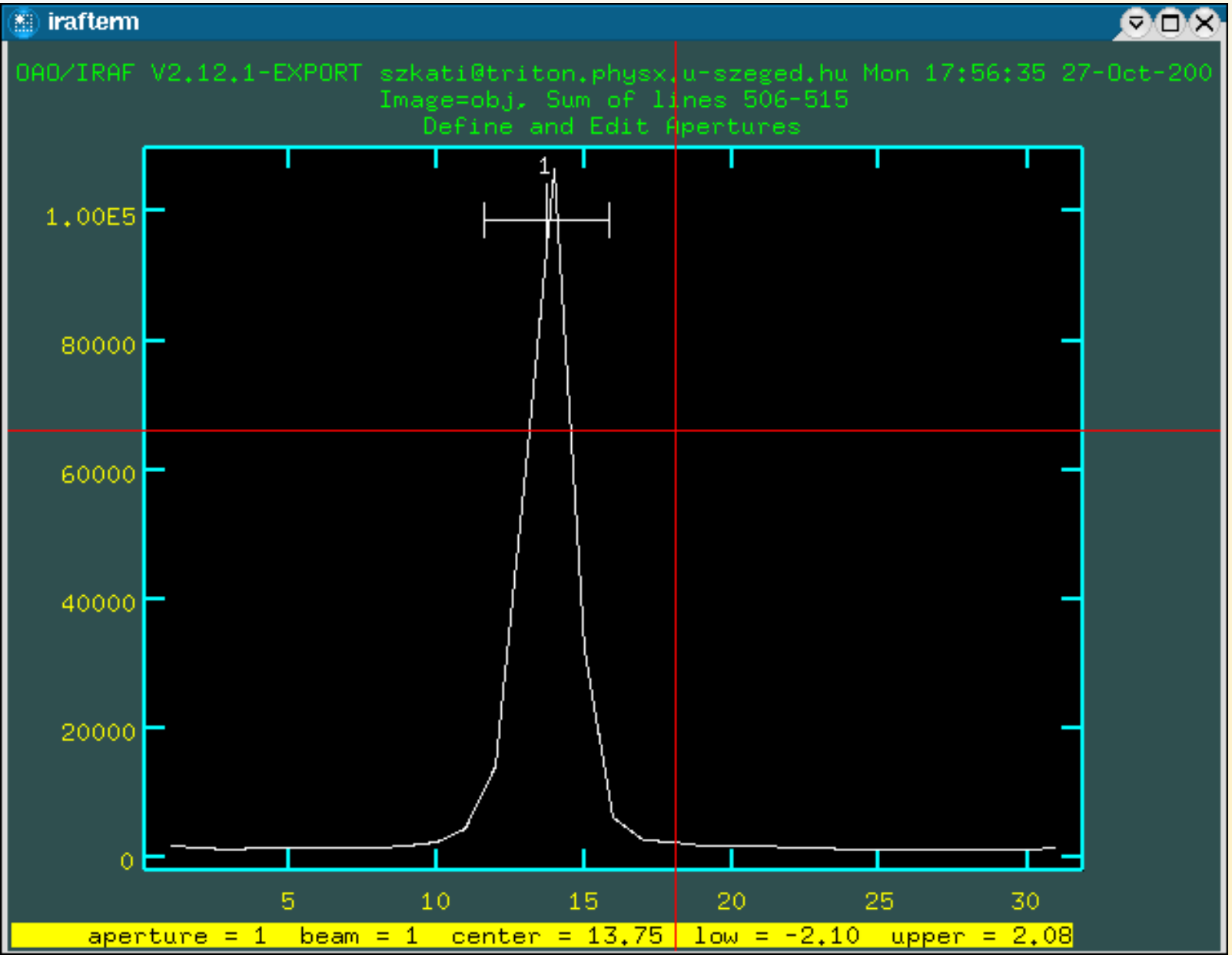
Extracting the aperture: apall

- noao => twodspec => apextract => epar apall



=> háttér illesztése

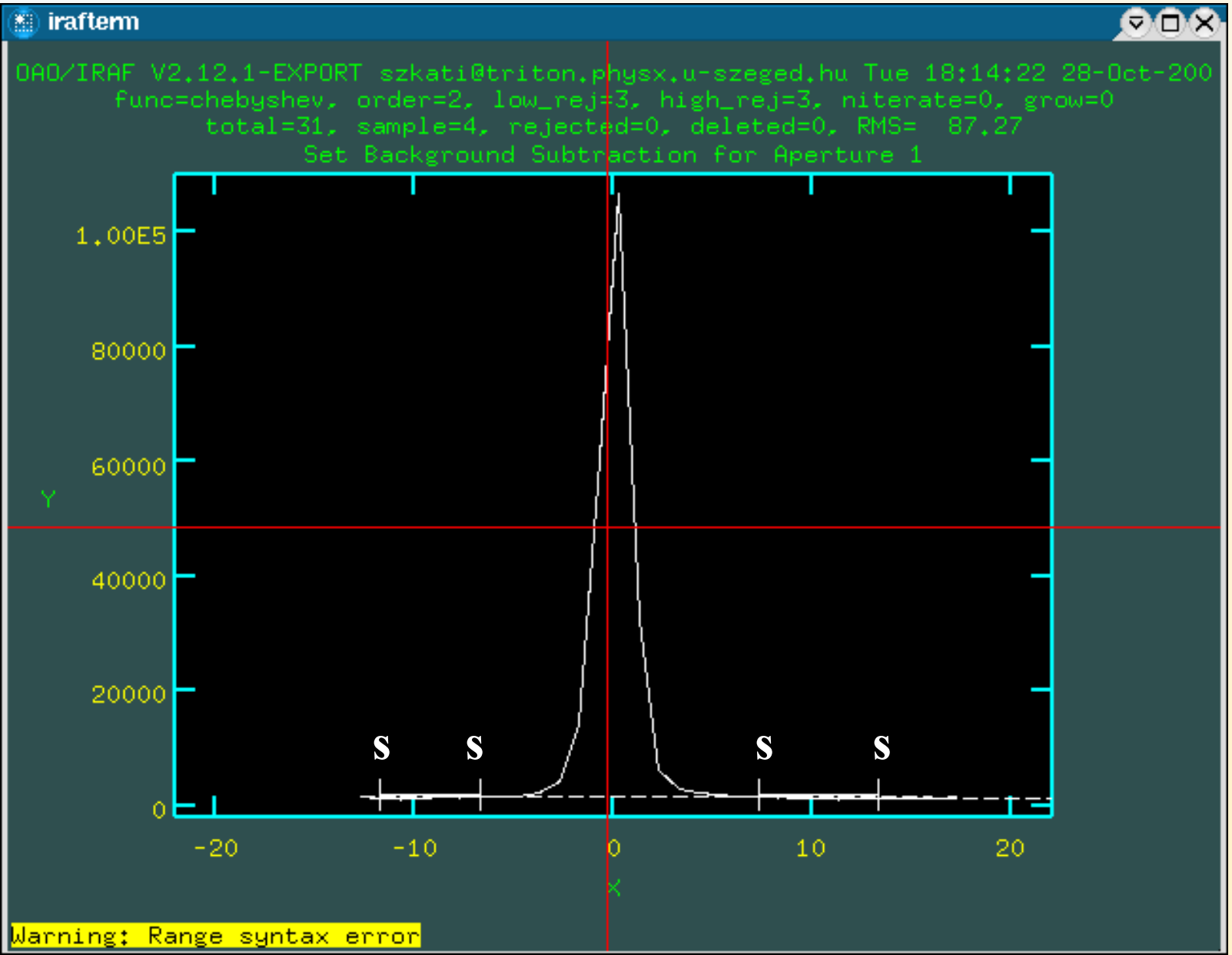
Extracting the aperture: defining the aperture



d : törlés
n : új apertúra poz.
l : bal oldal széle
u : jobb oldal széle

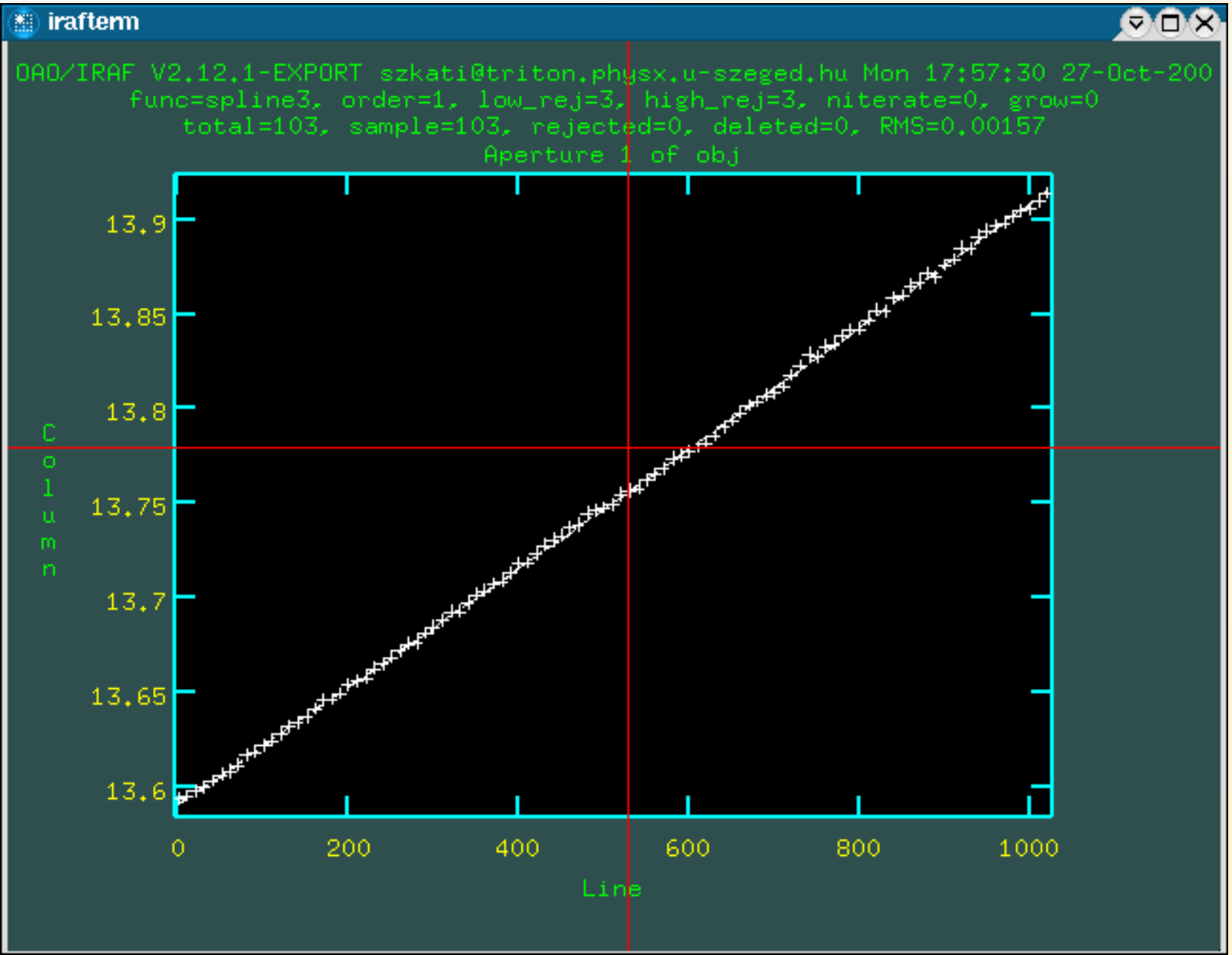
c: centrálás

Extracting the aperture: defining the background



t : háttér törlése
s : háttér kijelöl
f: illesztés
q : kilépés

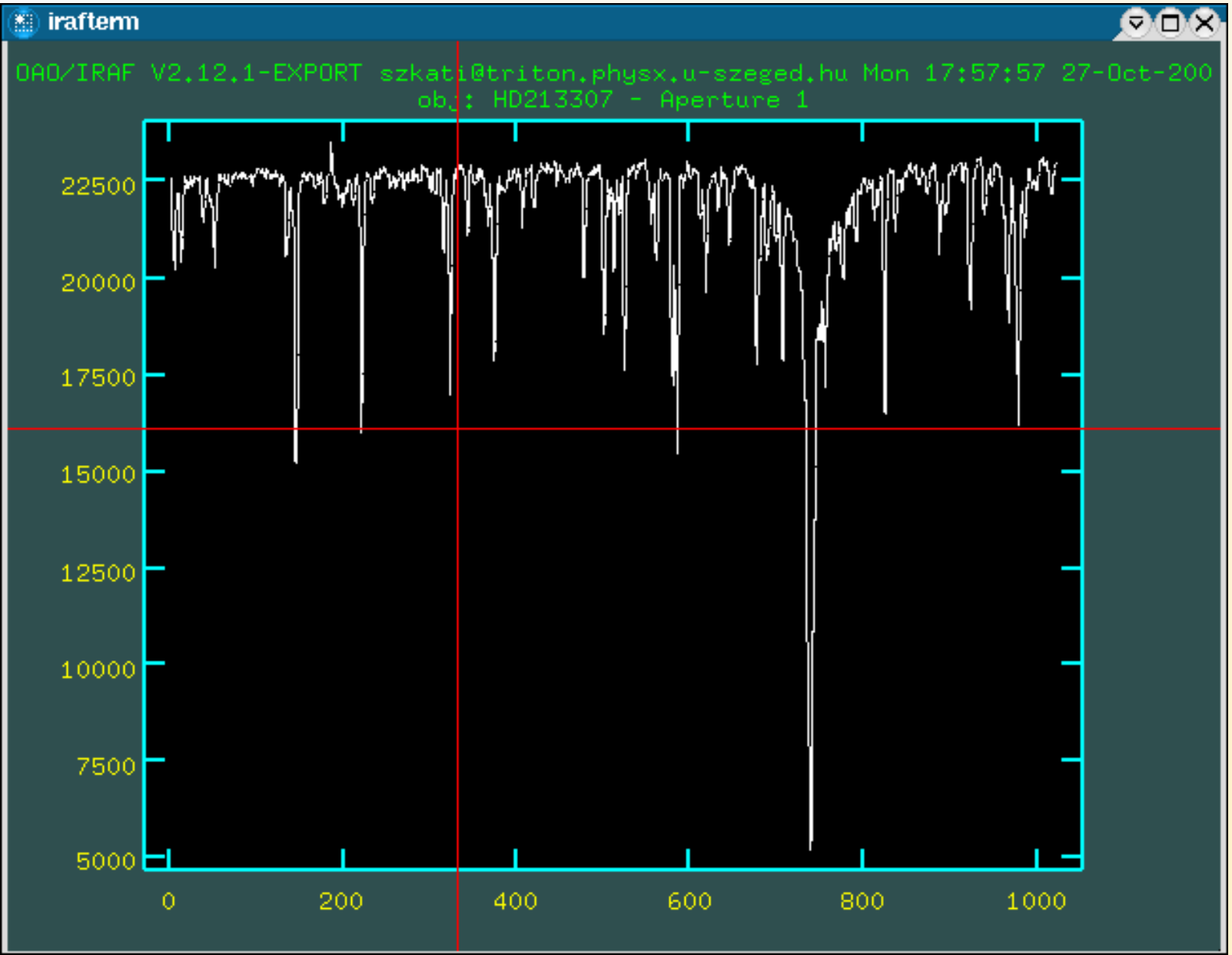
Extracting the aperture: tracing the aperture



Tracing the aperture by fitting a polynomial.

:order n rend megváltoztatása
:function legendre fgv. megváltoztatása
:niter 2 rossz pontok kiszórása

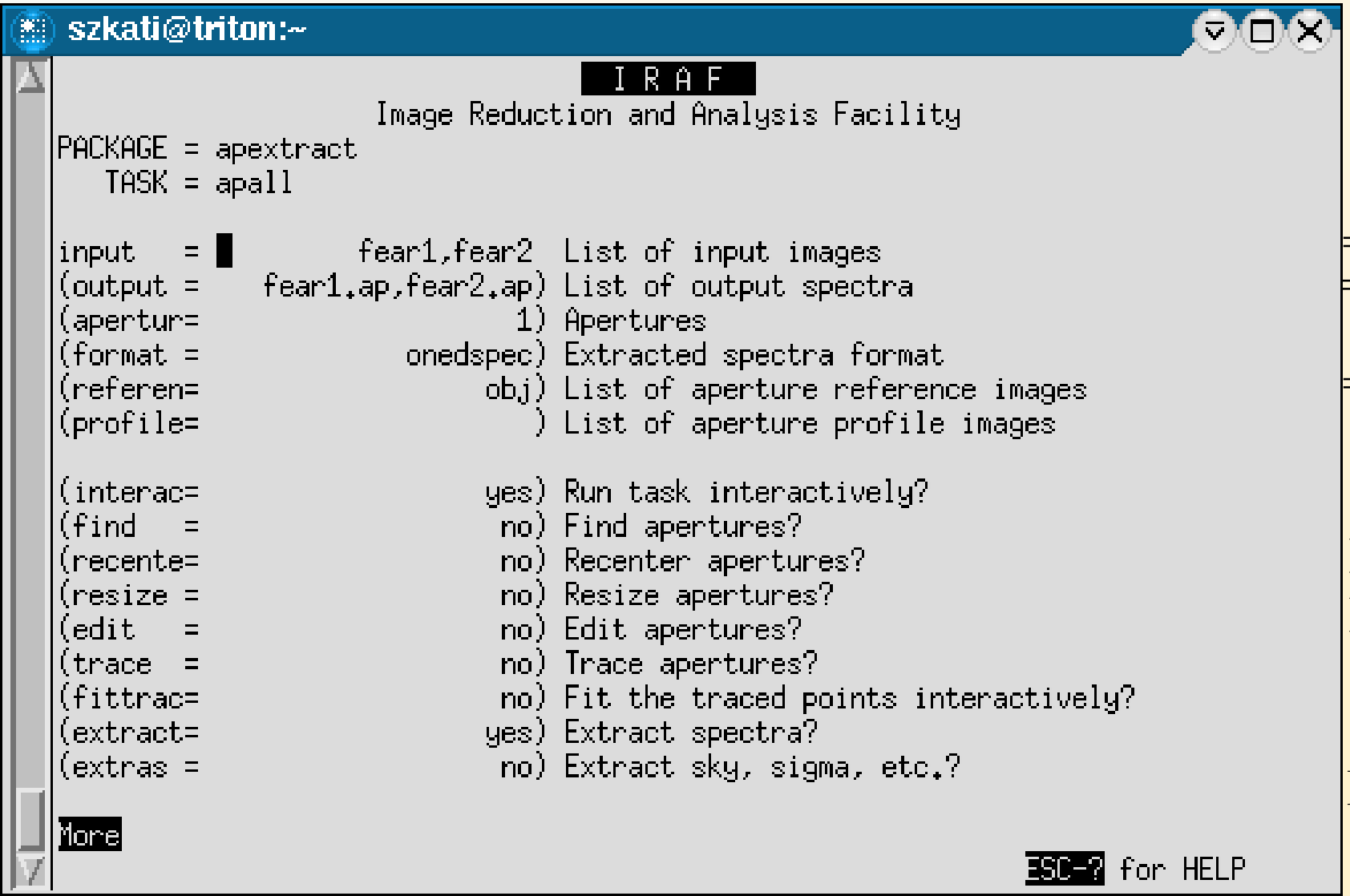
Extracting the aperture: the aperture



The traced spectrum, but it is not yet calibrated.

Extracting the aperture of the lamp: apall

- noao => twodspec => apextract => epar apall



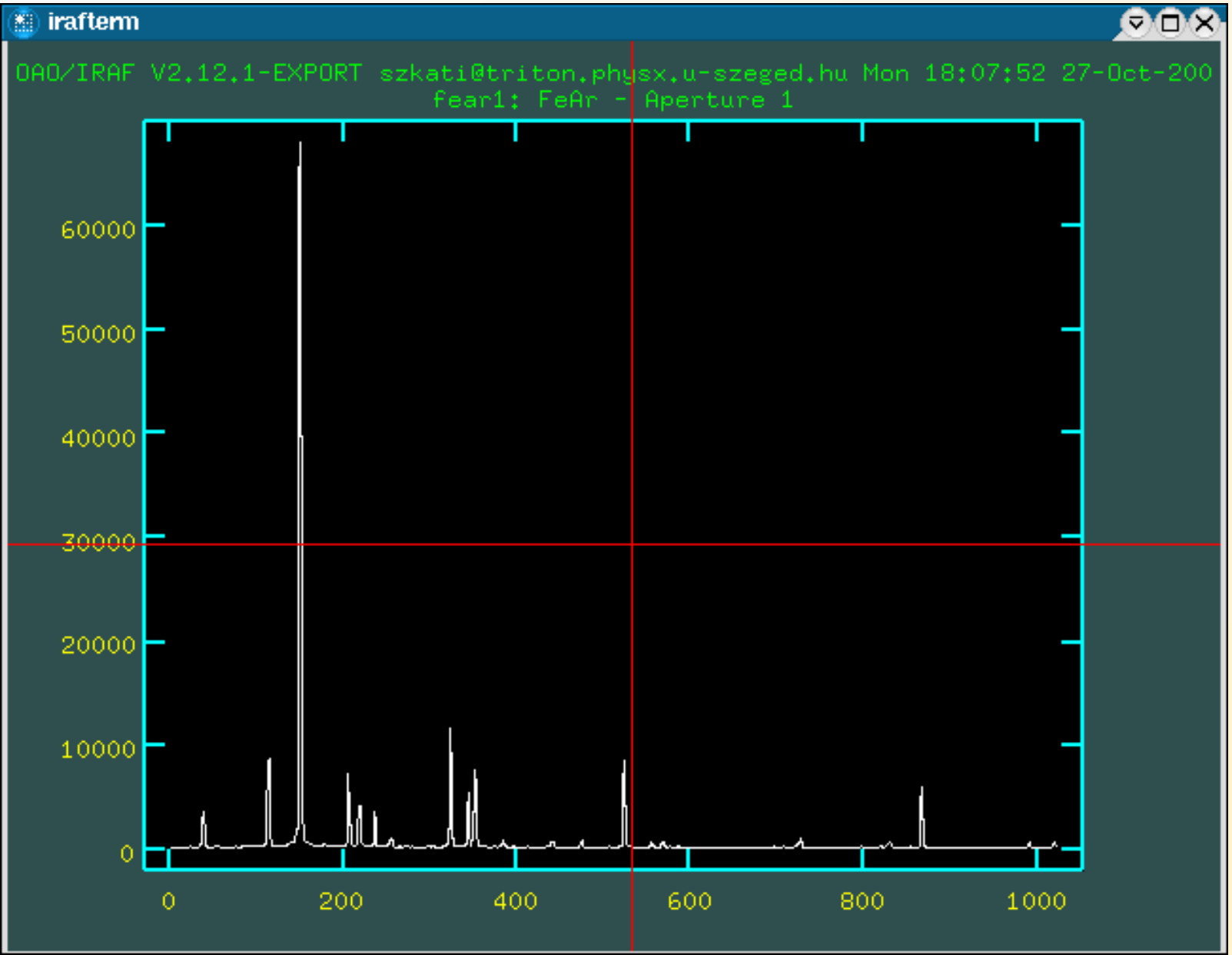
=> spektrállámpa spektrum
=> kimenő kép

=> objektum spektruma

Minden kérdésre a válasz: no
Kivéve: extras, interac

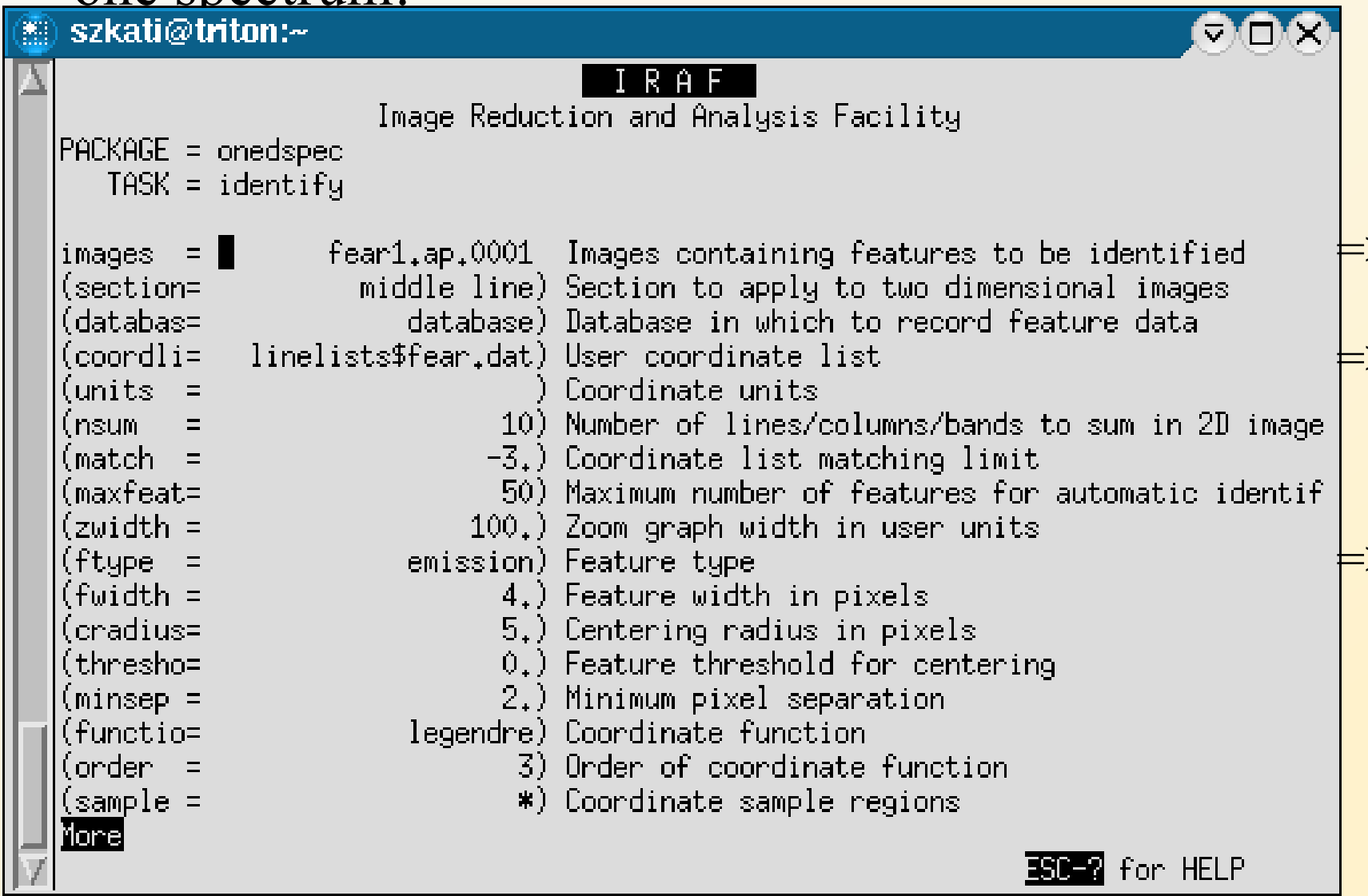
Background: none

Extracting the aperture: the lamp



Wavelength calibration: identify

- noao => onedspec => epar identify We only need to do this for one spectrum!



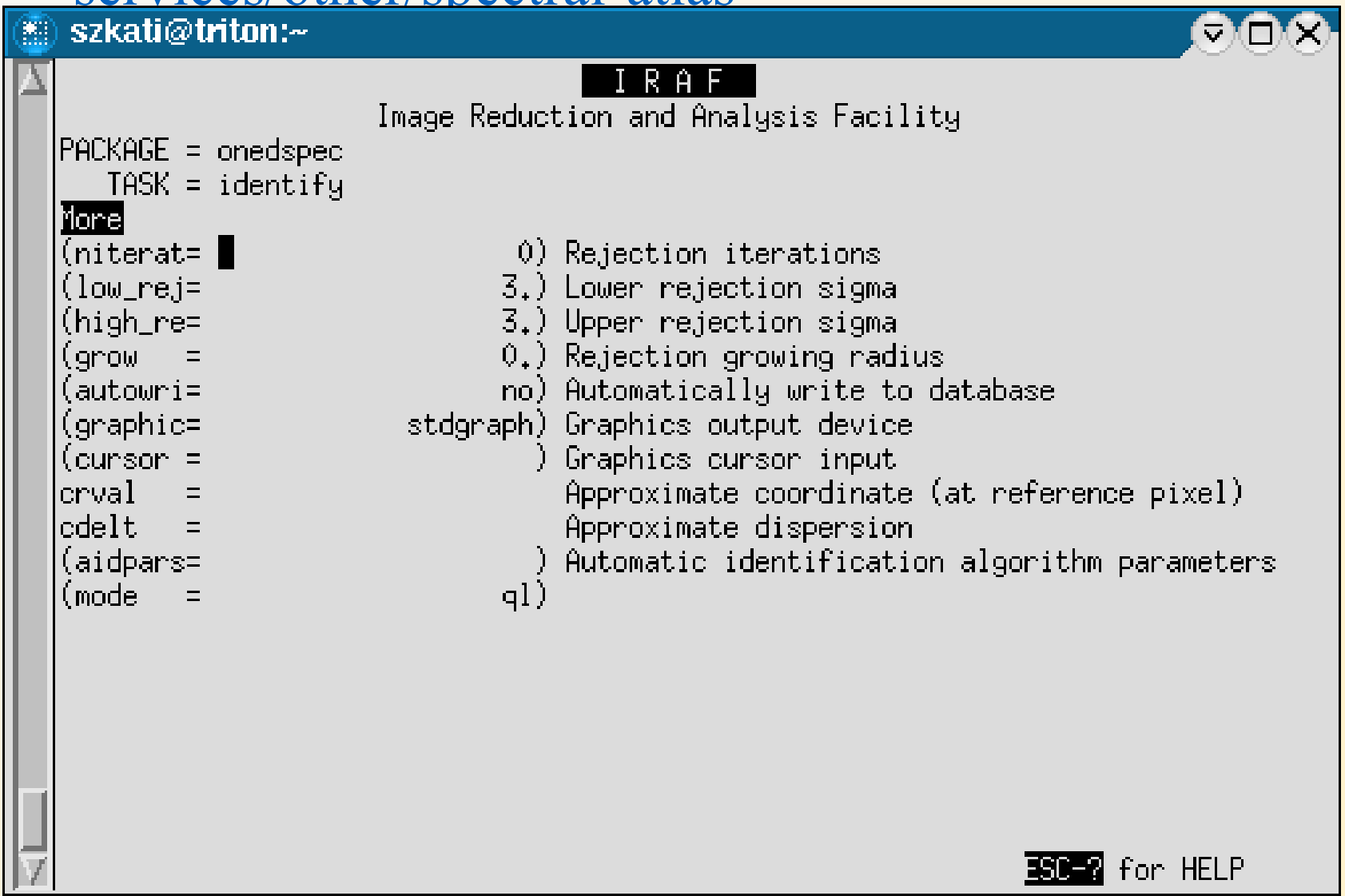
=>spektrállámpa spektrum

=> vonallista

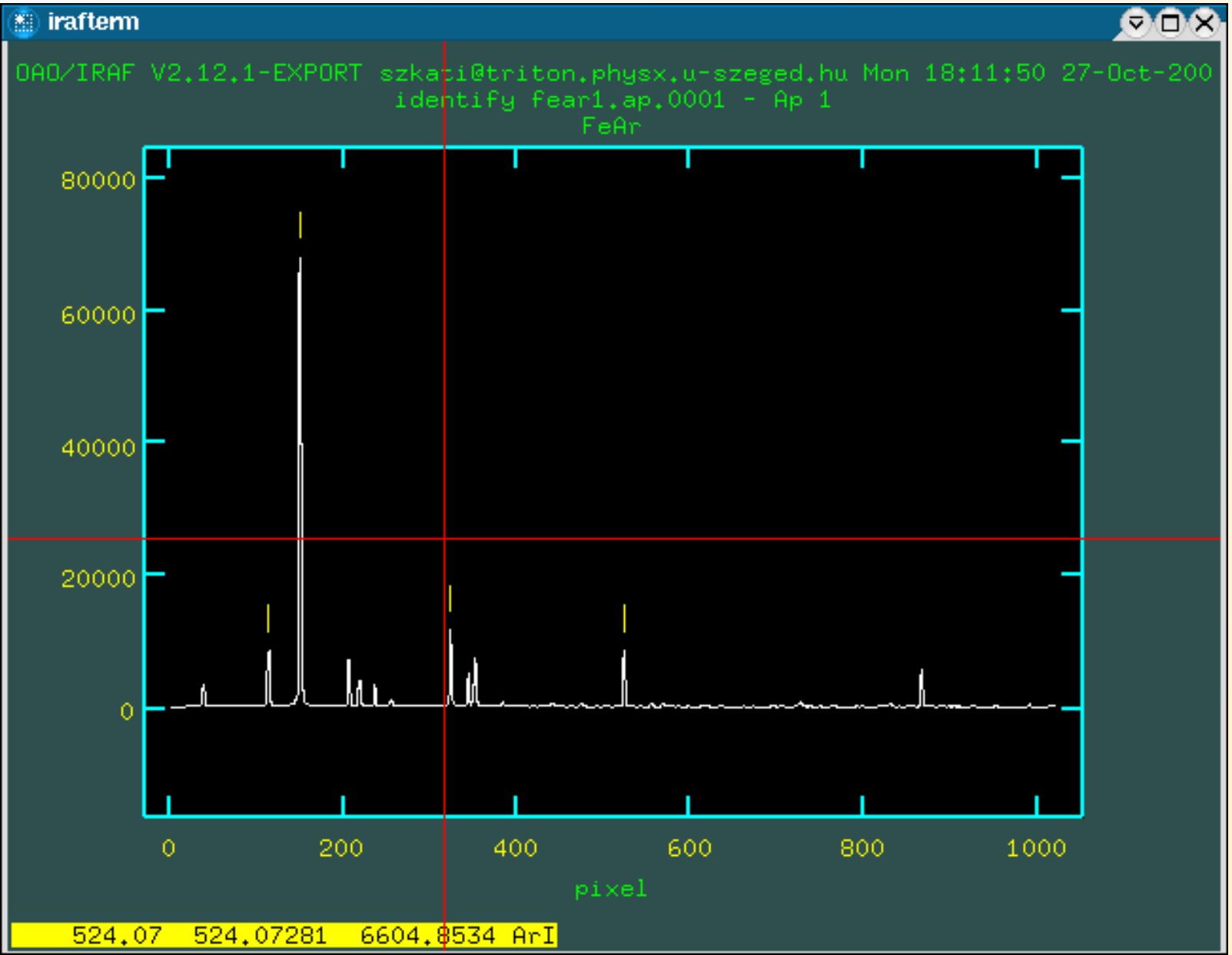
=> vonalak típusa

Wavelength calibration: identify

- noao => onedspec => epar identify https://noirlab.edu/science/data-services/other/spectral-atlas



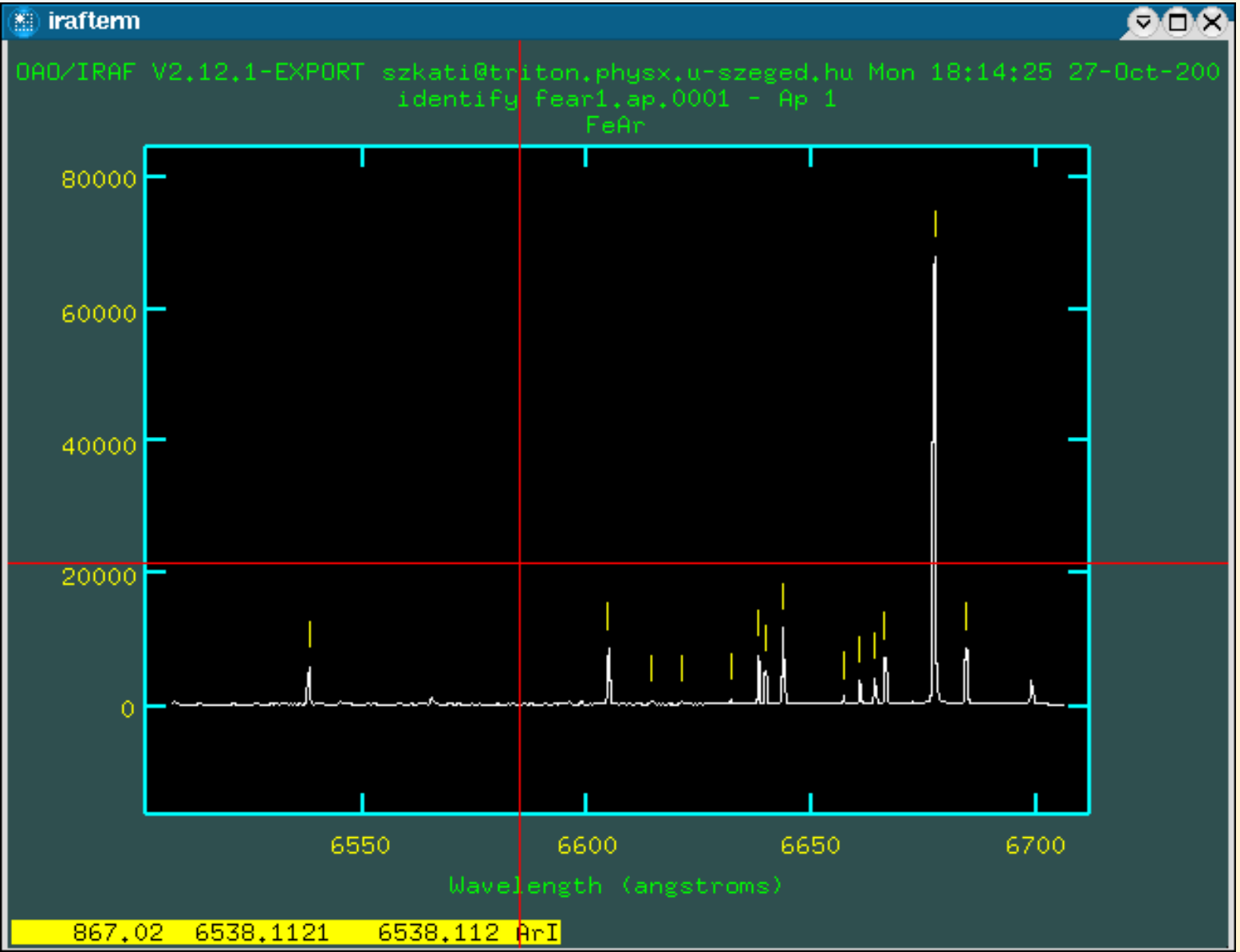
Wavelength calibration: identify



m => azonosítás

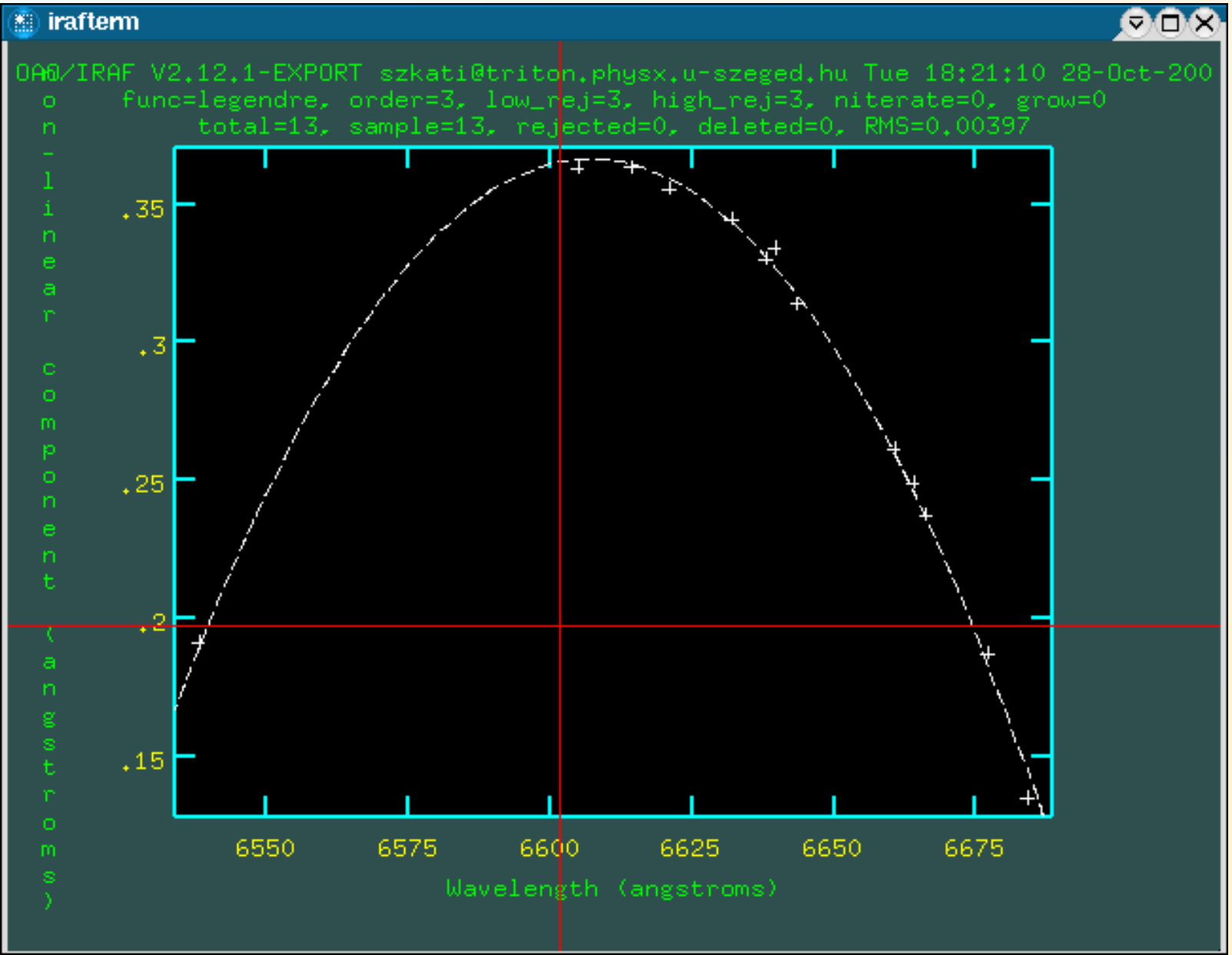
d => rontott törlése

Wavelength calibration: identify

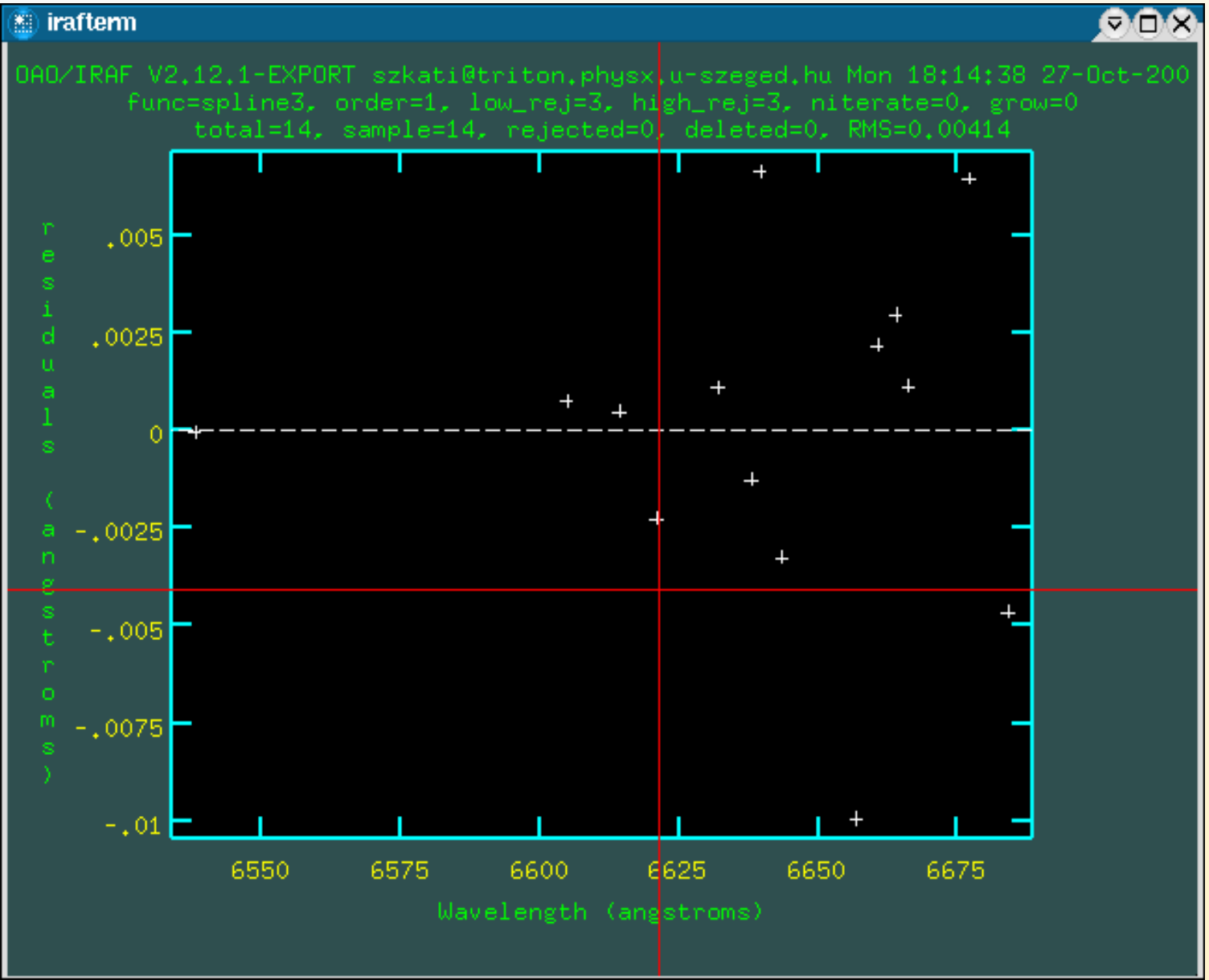


l => sok pont azonosítása
f => diszp.görbe illesztés

Wavelength calibration: identify



Wavelength calibration: identify

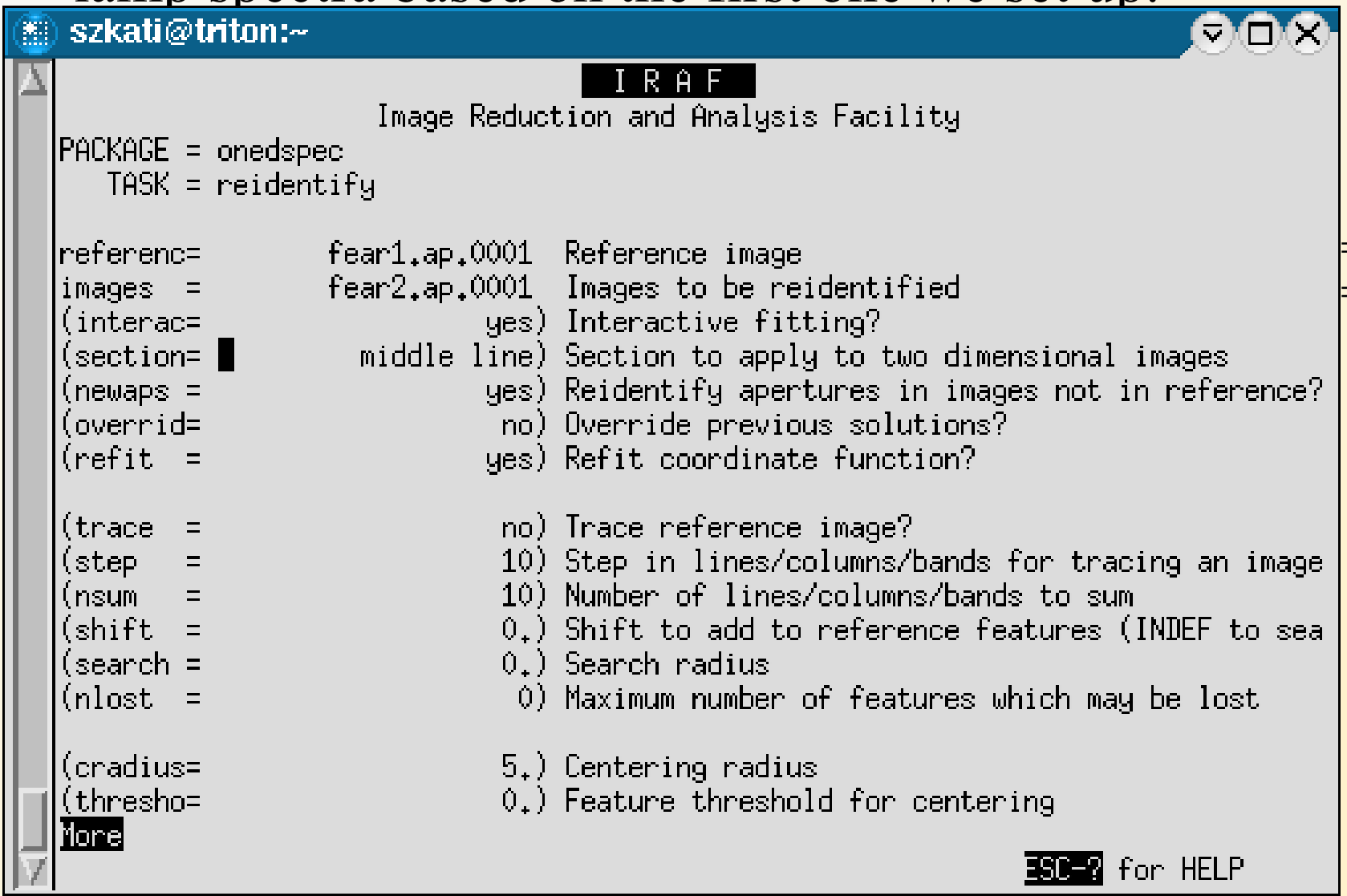


d => törli a pontot

f => új diszp. görbe illesztés

Wavelength calibration: reidentify

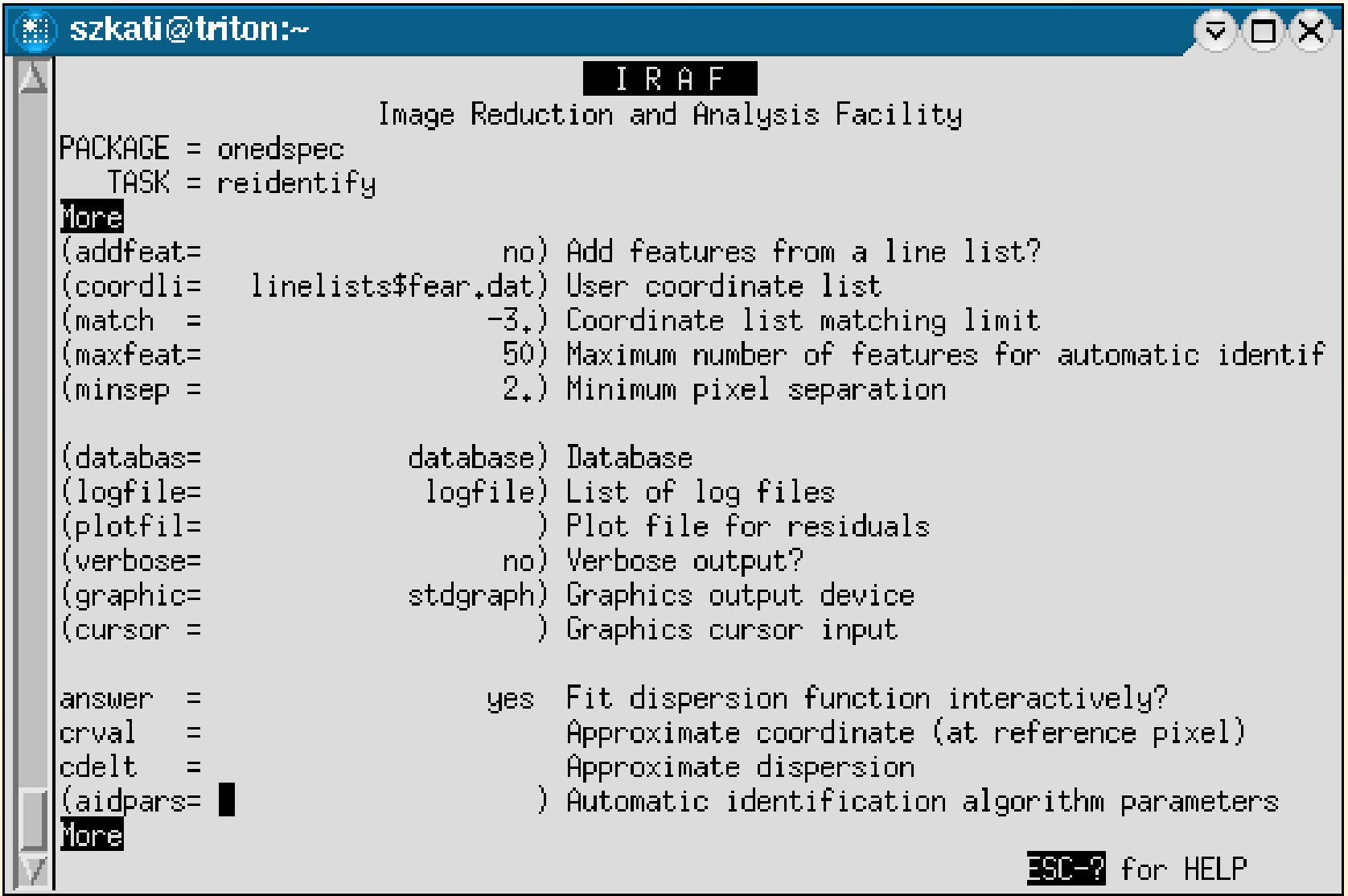
- noao => onedspec => epar reidentify It calibrates all other lamp spectra based on the first one we set up.



=> bekalibrált spektrum
=> összes spektrállámpa spektrum

Wavelength calibration: reidentify

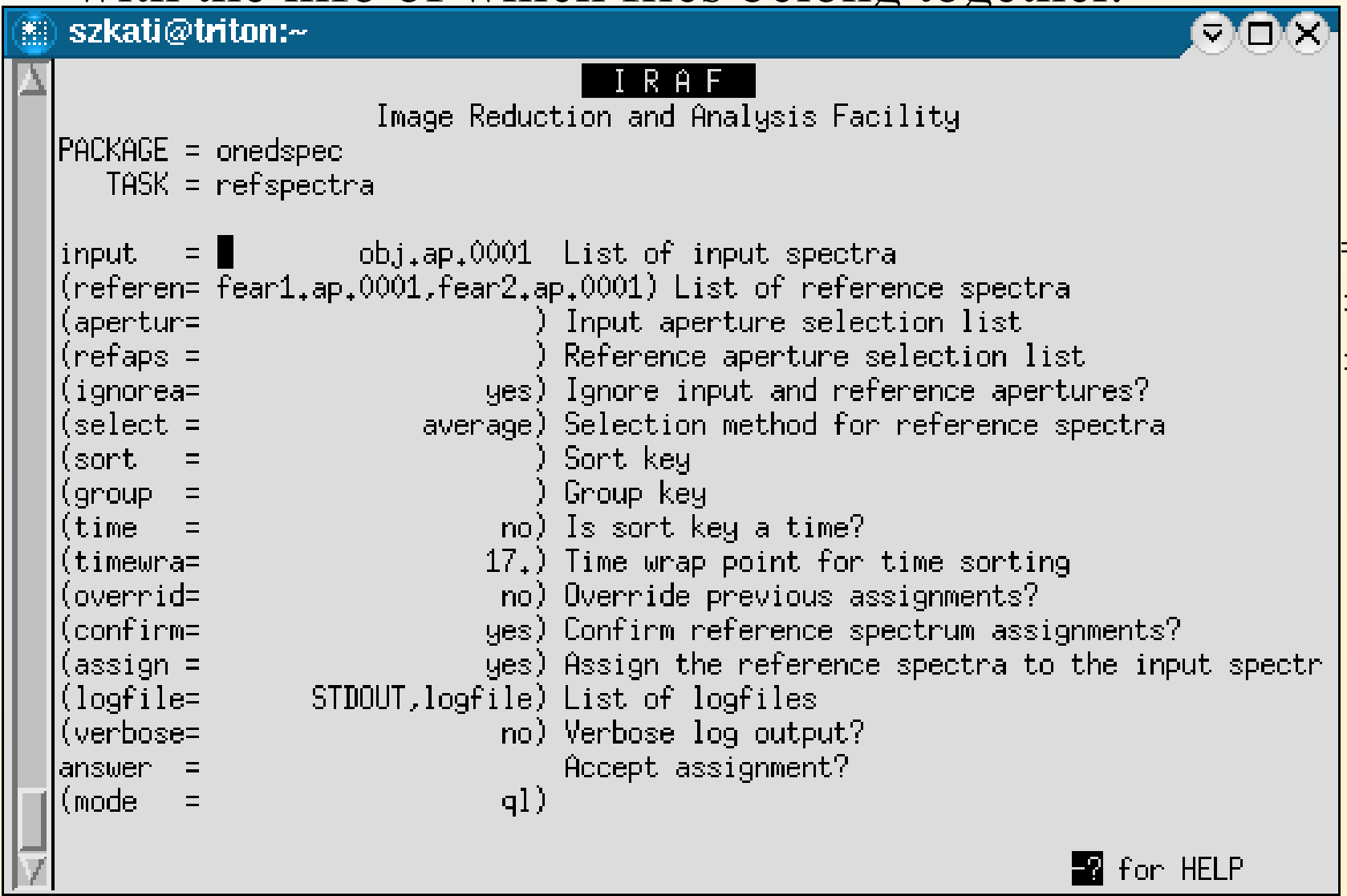
- noao => onedspec => epar reidentify



Futatás közben:
q =>kilépés;

Wavelength calibration: refspectra

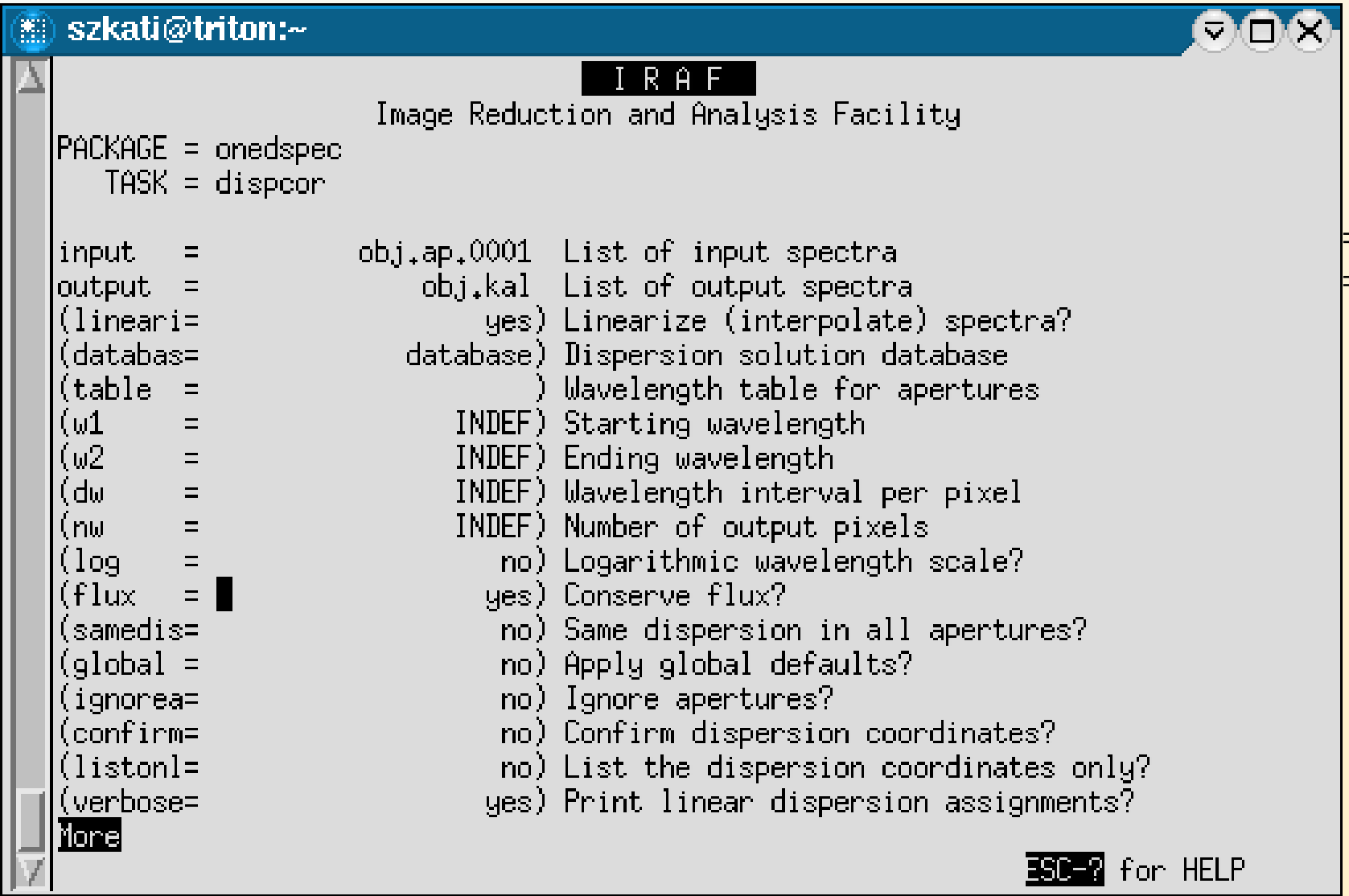
- noao => onedspec => epar refspectra it will edit the header with the info of which files belong together.



=> csillag spektruma
⇒ a spektrumhoz tartozó
⇒ spektrállámpa spektrumok

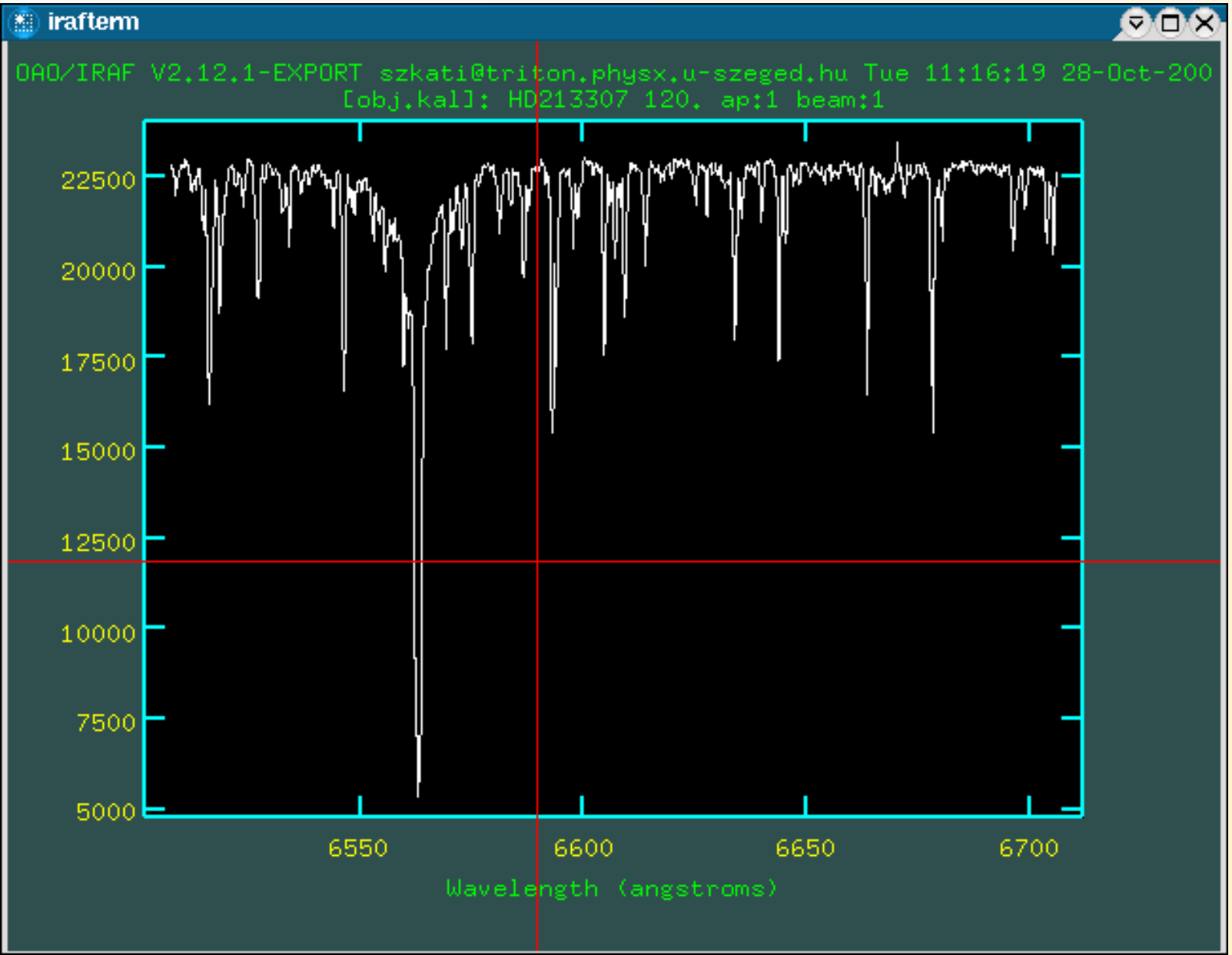
Wavelength calibration: dispcor

- noao => onedspec => epar dispcor calibrating the object's spectrum.



=> bemenő file
=> kimenő file

Wavelength calibration: splot

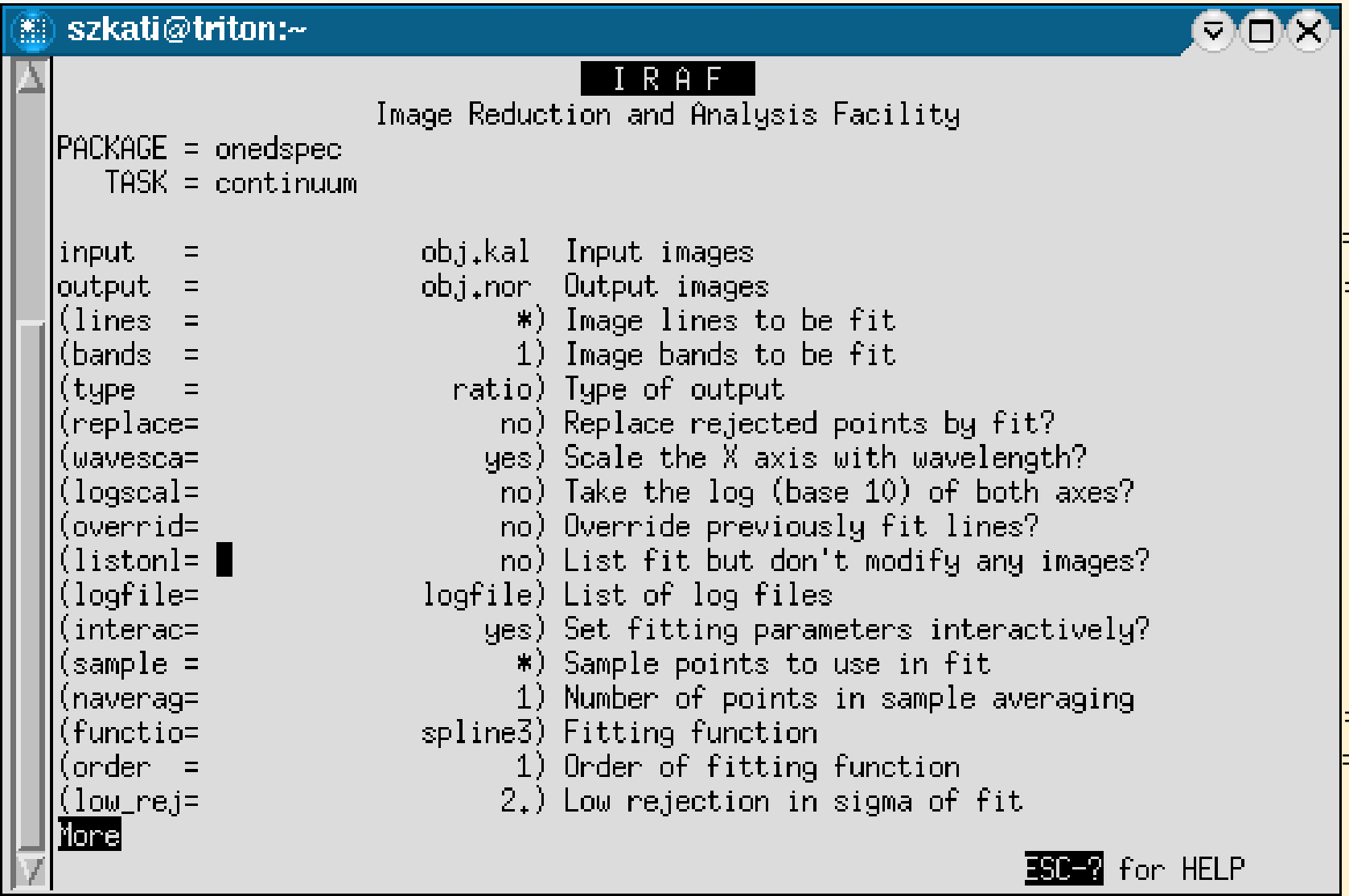


Continuum normalization: continuum

- Continuum normalization: a process in which we fit the continuum of a star's spectrum and then divide the entire spectrum with the fitted spectrum.
- The normalized fluxes will change between 0 and 1.
- Continuum normalization is not about subtracting the continuum!
- Use continuum normalization: when determining atmospheric parameters, abundances, measuring equivalent widths, and radial velocity of stars. While radial velocity is less sensitive to the normalization, accurate parameters and abundances can only be derived when the continuum normalization is excellent.
- Use continuum subtraction: to remove sky and telluric lines from the observation, to remove the spectrum of a background galaxy/cloud.
- The continuum subtracted spectra are often continuum normalized later.

Continuum normalization: continuum

- noao => onedspec => epar continuum normalizing the continuum.

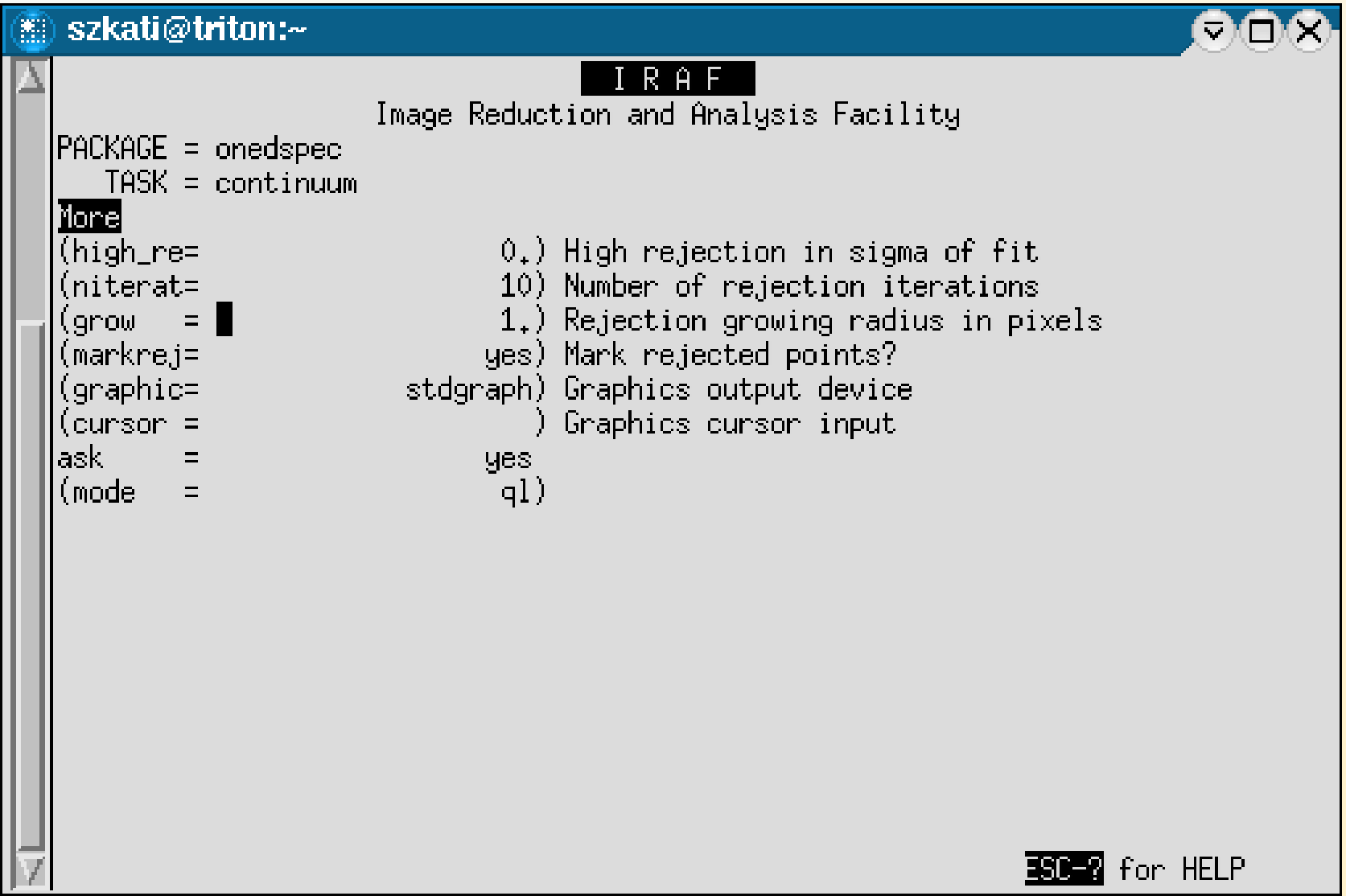


=> bemenő file
=> kimenő file

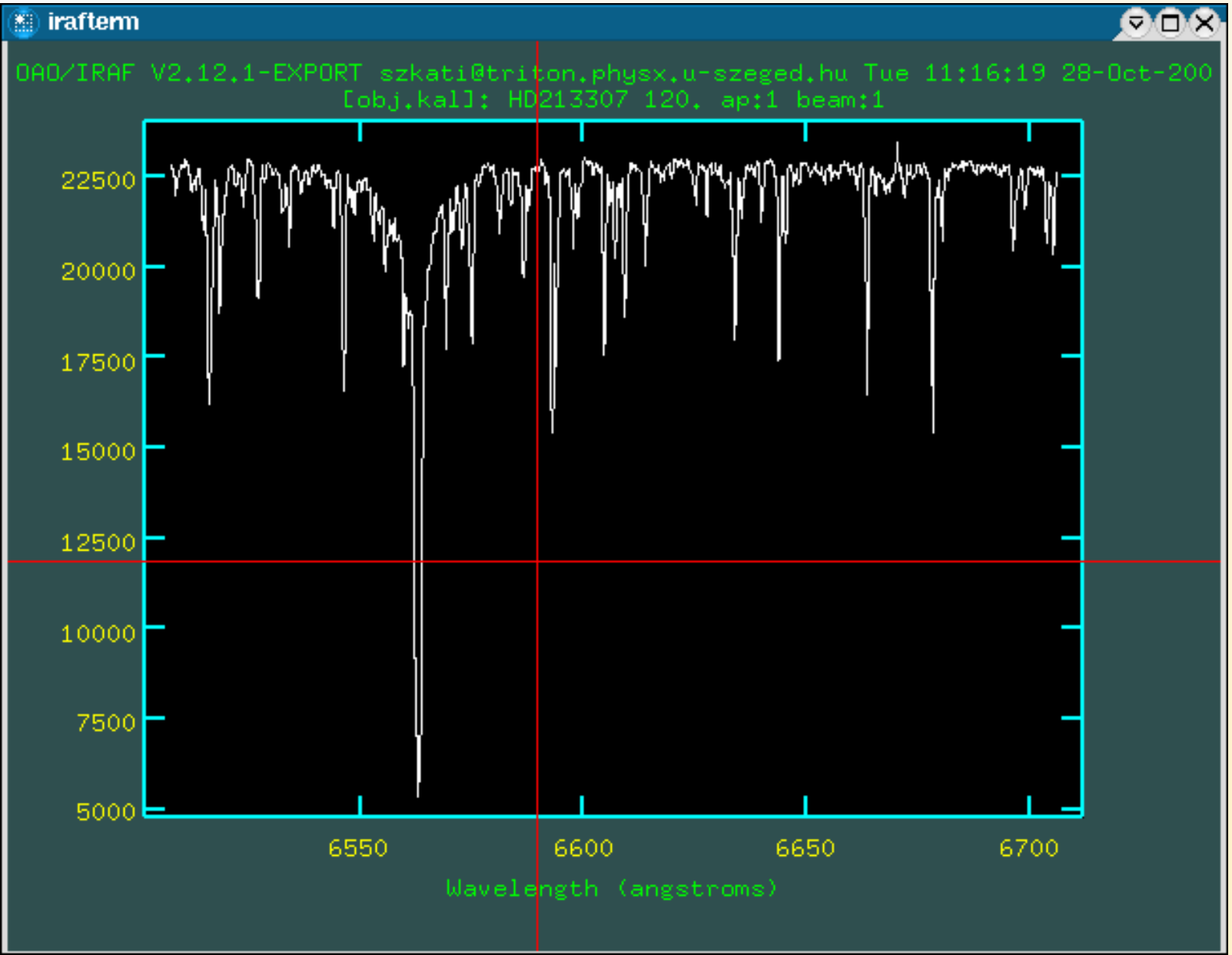
=> függvény
=>illesztés rendje

Continuum normalization : continuum

- noao => onedspec => epar continuum

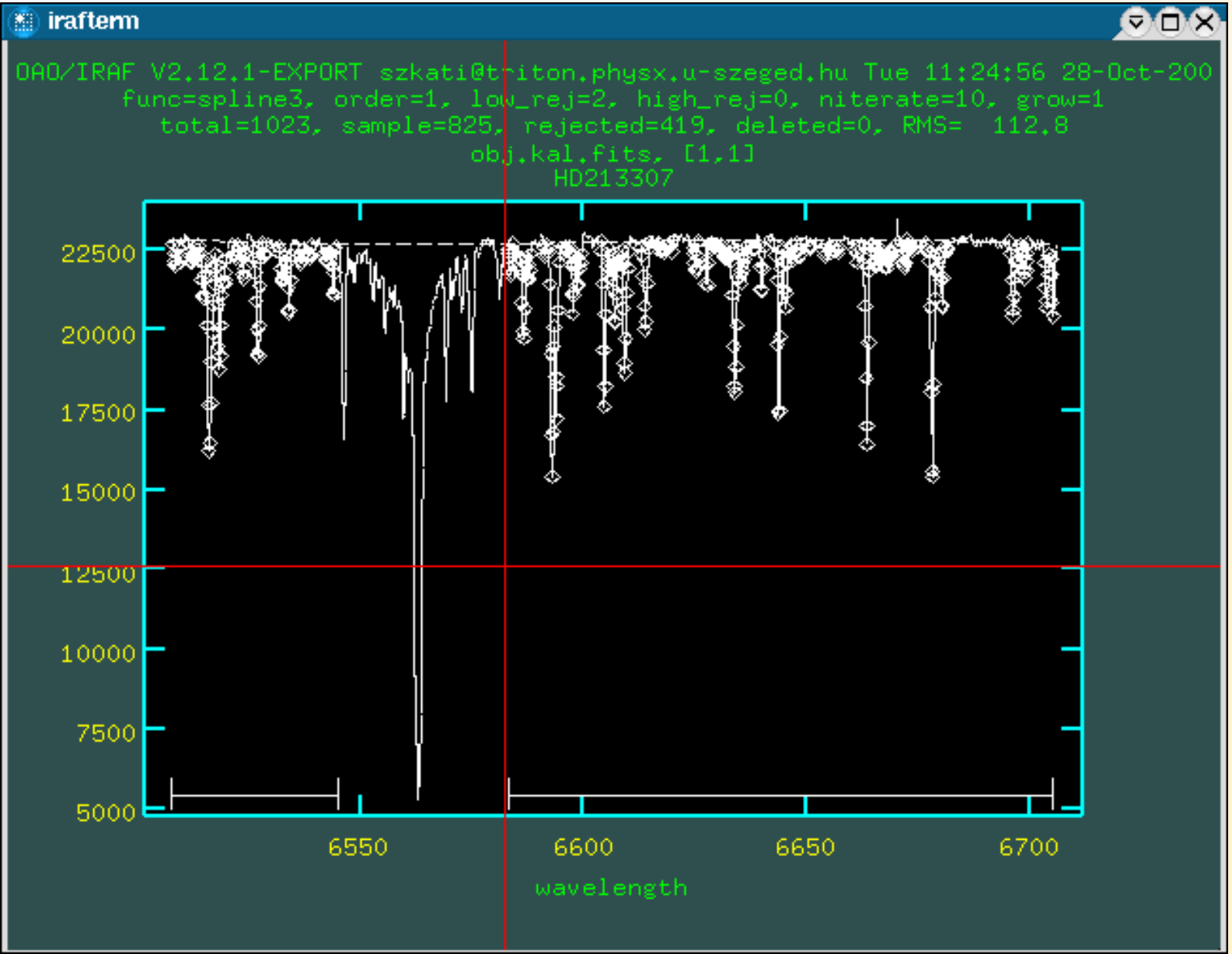


Continuum normalization: continuum



Remove wide lines in order to improve accuracy.

Continuum normalization: continuum



2xs => határok kijelőlése
f => új illesztés

The reduced spectrum!

